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SIEMENS

LSafe Vote&Scale Library for TIA Portal STEP 7

STEP 7 Basic/Professional V18 (TIA PORTAL)

<https://support.industry.siemens.com/cs/ww/en/view/109826840>

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1 Overview of the library

1.1 General

The “LSafe Vote&Scale” library contains fail-safe blocks for the implementation of various safety applications using F-CPU's of the SIMATIC S7-1200 or S7-1500 series together with fail-safe I/O and SINAMICS converters via PROFIsafe.

These building blocks are provided in the form of a library and can be used freely. The finished functions are freely adaptable and can therefore be used without restrictions.

The library described here is versioned and will be continuously expanded. Information on versioning can be found in the Appendix, Chapter 5 under “Change History”.

Note The library described is TÜV-certified. The TÜV report is only valid for the current version, the blocks specified in the report and their signature (CRC).

Modification of the library blocks leads to a change in the block signature (CRC). If the signature (CRC) differs, the validity of the TÜV report can no longer be referenced and a new evaluation is required

Note The integrated safety functions of SINAMICS are certified according to SIL2 according to IEC 61508 or IEC 61800-5-2, as well as PL d and category 3 according to ISO 13849-1.

The PROFIsafe communication and the fail-safe PLC are certified according to SIL3 according to IEC 61508 or EN 62061, as well as PL e and category 4 according to ISO 13849-1.

Note Certain faults can only be detected by a signal change (e.g. welded contactor). Make sure that you select an appropriate diagnostic test interval for your safety function.

1.2 Hardware and software requirements

Requirements for this library

To be able to use the functionality of the library described here, the following hardware and software requirements must be met.

Hardware Requirements

All blocks (FB, FC, DB, ...) in the library can be used without restriction with the following controllers:

- SIMATIC S7-1200F product family (from firmware V4.2)
- SIMATIC S7-1500F product family (from firmware V2.6)
- Simulation with SIMATIC S7-PLCSIM (from V18)

The environmental conditions are derived from the standard environmental conditions for the above hardware portfolio and can be found in the corresponding manuals.

Software Requirements

- SIMATIC STEP 7 Basic/Professional (TIA Portal V18 or higher)
- SIMATIC STEP 7 Safety Advanced V18 or higher

Note

It is generally possible to open a library with STEP 7 Basic even though STEP 7 Professional elements (e.g. SIMATIC S7-1500 controller) are included. In this case, you will be informed by a message when you open the library.

All elements (types and copy templates) can be used if they are supported by the installed hardware in the TIA Portal.

If you want to copy elements with STEP 7 Basic from the library that are not supported (for example, SIMATIC S7-1500 controller), an error message is displayed.

2 Working with the library

2.1 General

All blocks in the “LSafe Vote&Scale” library can be freely used in conjunction with SIMATIC S7-1200F and SIMATIC S7-1500F controllers.

The blocks are stored as a type in the library. This allows you to benefit from the following advantages:

- Central update function of library items
- Versioning of library elements

Note

Information on general library handling can be found in the library handling guide <https://support.industry.siemens.com/cs/ww/en/view/109747503> and in the programming guide for S7-1200/1500 in the “Libraries” chapter <https://support.industry.siemens.com/cs/ww/en/view/81318674>

More information on libraries in the TIA Portal:

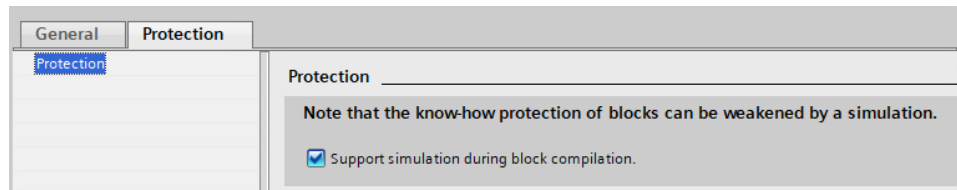
- Libraries topic page
<https://support.industry.siemens.com/cs/ww/en/view/109738702>
- How do you open libraries in STEP 7 (TIA Portal)?
<https://support.industry.siemens.com/cs/ww/en/view/37364723>
- Automate in less than 10 minutes TIA Portal: Time Savers - Global Libraries
<https://support.industry.siemens.com/cs/ww/en/view/78529894>
- Which elements from STEP 7 (TIA Portal) can be stored in a library as a type or as a master copy?
<https://support.industry.siemens.com/cs/ww/en/view/109476862>
- How can you automatically open a global library when starting TIA Portal V13 and higher and use it as a company library, for example?
<https://support.industry.siemens.com/cs/ww/en/view/100451450>
- Library with PLC data types (LPD) for STEP 7 (TIA Portal) and SIMATIC S7-1200 / S7-1500
<https://support.industry.siemens.com/cs/ww/en/view/109482396>

2.2 Simulatability with SIMATIC S7-PLCSIM Advanced

The simulation with SIMATIC S7-PLCSIM Advanced is deactivated in the properties of the library.

To ensure that the function blocks can be simulated after compilation with SIMATIC S7-PLCSIM Advanced, proceed as follows:

1. Open the properties of your project (Select the project in the project navigation, open the context menu by clicking on the right mouse button [alternatively the second mouse button] and select "Properties..." off)
2. In the "Protection" tab, activate the option "Support simulation during block compilation".



Note

Blocks with activated simulation capability occupy more storage space in the PLC.

3 Structure and design of the library

The following chapters describe all building blocks of the library “LSafe Vote&Scale”.

The chapters are structured in the same way as the library itself.

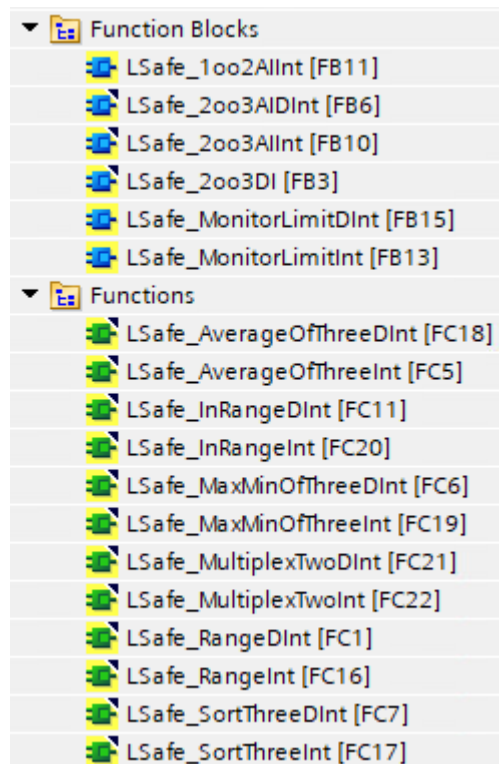


Table: Overview of Signatures of Function blocks

Name	Signature	Version
LSafe_1oo2AlInt	0xF489D46	V1.0
LSafe_2oo3AlDInt	0xE2CF83D4	V1.0
LSafe_2oo3AlInt	0x92B4A669	V1.0
LSafe_2oo3DI	0x14860189	V1.0
LSafe_MonitorLimitDInt	0x2391F2B3	V1.0
LSafe_MonitorLimitInt	0xB29C5598	V1.0

Table: Overview of Signatures of Functions

Name	Signature	Version
LSafe_AverageOfThreeDInt	0xAF5126FA	V1.0
LSafe_AverageOfThreeInt	0x3A6D2F44	V1.0
LSafe_InRangeDInt	0x1DC622CA	V1.0
LSafe_InRangeInt	0xF88B1391	V1.0
LSafe_MaxMinOfThreeDInt	0x9DCD5CE3	V1.0
LSafe_MaxMinOfThreeInt	0x5ED989CE	V1.0
LSafe_MultiplexTwoDInt	0xF8AB2E4D	V1.0
LSafe_MultiplexTwoInt	0x41E99CE6	V1.0

3 Structure and design of the library

Name	Signature	Version
LSafe_RangeDInt	0x56534873	V1.0
LSafe_RangeInt	0xB8092148	V1.0
LSafe_SortThreeDInt	0x8C35FA73	V1.0
LSafe_SortThreeInt	0x74A4AA8	V1.0

Note

The function of the mentioned devices is tested according to SIL 3 (IEC 62061) or PLe (ISO 13849-1). The analysis applies only to the signatures mentioned in the tables: "Overview: Signatures of the function blocks" and "Table: Overview of Signatures of Functions". If the function modules are changed, the assumptions of the safety assessment because a correct function can no longer be ensured. A change can be detected by the signature of the blocks.

The function blocks may only be operated up to the specified SIL according to IEC 62061 or PL according to 13849-1.

4 Program blocks

4.1 Function Blocks

4.1.1 LSafe_1oo2AIInt (SafetyFB / V1.0.0)

Author: Siemens

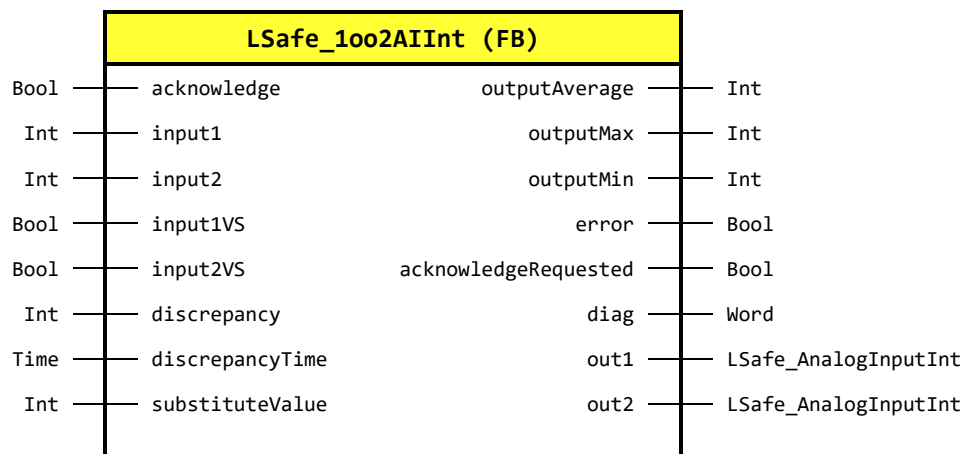
Short description

The block LSafe_1oo2AIInt evaluates two analog input signals and uses them to determine the arithmetic mean, the minimum and maximum value. For at least one invalid value, the substitute value will be returned. The block monitors whether the analog input values within the specified discrepancy time are within the specified discrepancy to each other. If this is not the case for an input value, it is deactivated until it is acknowledged. The output error is activated, if at least one of both values becomes invalid. This can happen for two reasons,

1. first, if the value status is invalid
2. if the value is not within the discrepancy and the discrepancy time expires.

Interface description

Block interface



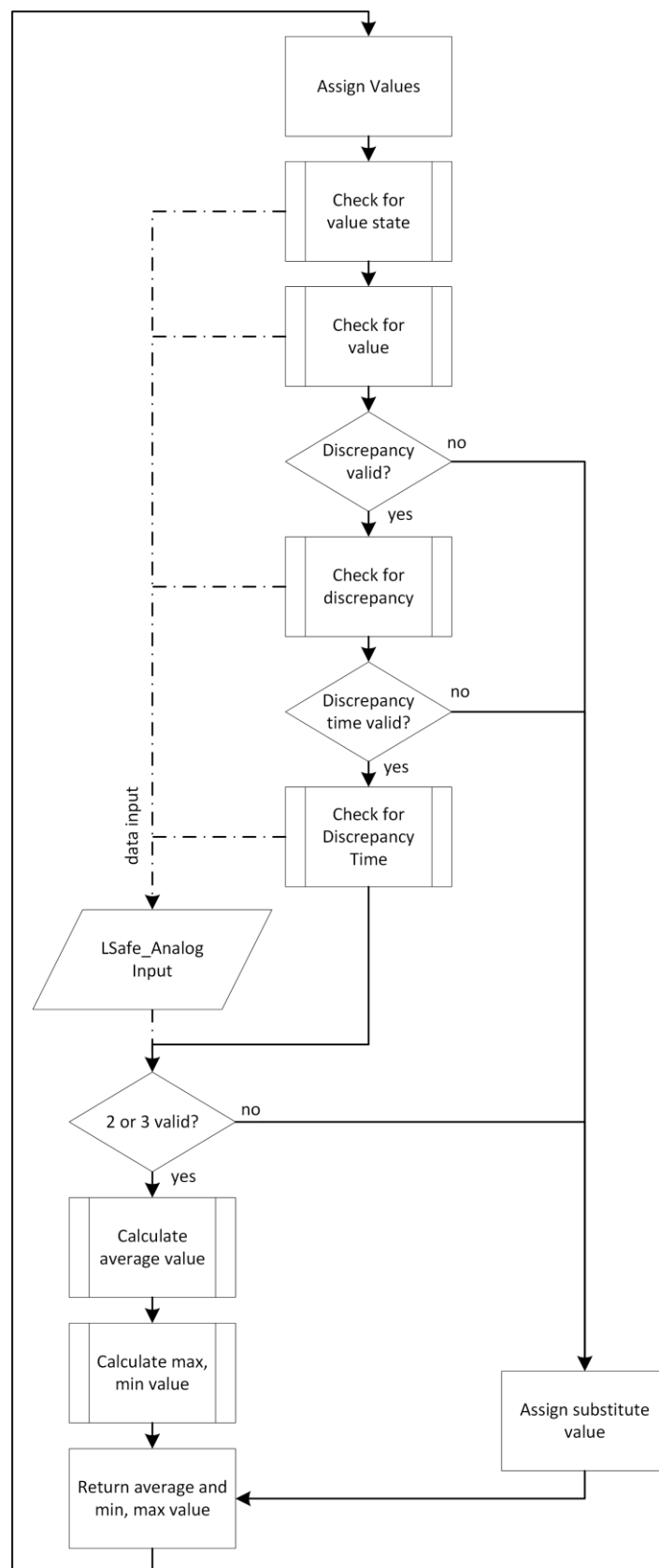
Input parameter

Identifier	Data type	Default value	Description
acknowledge	Bool	FALSE	Acknowledge, positive edge signal, NO
input1	Int	0	Input Value 1
input2	Int	0	Input Value 2
input1VS	Bool	FALSE	Value State of Input Value 1, False: fault
input2VS	Bool	FALSE	Value State of Input Value 2, False: fault
discrepancy	Int	0	Discrepancy maximum between two values
discrepancyTime	Time	T#0s	Discrepancy time, longest time for values to reach discrepancy
substituteValue	Int	0	Substitute value, if fault outputAverage, outputMax and outputMin will return substitute value

Output parameter

Identifier	Data type	Description
outputAverage	Int	Output of arithmetic mean. Returns substitute value if less than two inputs are valid.
outputMax	Int	Outputs the maximum value of two valid inputs.
outputMin	Int	Outputs the minimum value of two valid inputs.
error	Bool	True, if at least one value properties invalid.
acknowledgeRequested	Bool	True, if at least one value properties can be acknowledged. Acknowledging do not lead implicitly to valid outputs.
diag	Word	DIAG output.
out1	LSafe_AnalogInputInt	Output of Input 1
out2	LSafe_AnalogInputInt	Output of Input 2

Functional description



NOTICE To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE The enable input (EN) must not be wired for F-modules.



WARNING

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Internal block description

In the first step, the block assigns substitute values to the internal variables and the external values.

In the following networks, the block checks various parameters of the analog input values. Furthermore, the attached discrepancy and discrepancy time are checked for validity.

In the second step, the block checks the value statuses and sets the input values to invalid if the value status is invalid.

In the third step, the input variables are checked for validity. They are checked for a range between or equal to MIN_VALUE and MAX_VALUE. The corresponding values are stored in the data type.

In the fourth step, the block checks the discrepancy for validity. If the input is invalid, the outputs are set to the equivalent values. The discrepancy may be between MIN_DISCREPANCY and MAX_DISCREPANCY.

In the fifth step, the block checks the variables for discrepancy. In this case, two variables may be separated at most in the specified discrepancy. The results are stored in the data type.

In the sixth step, the specified discrepancy time is checked for a maximum of MAX_DISCREPANCY_TIME (60s). If the value is invalid, the procedure is equivalent to the discrepancies that are invalid.

In the seventh step, timers are started when there is a discrepancy. If the input variables are not within the discrepancy within the discrepancy time, they are set to invalid and this information is cached in the data type.

In step 8 of the block, the input properties are checked for validity and correspond to the number of valid variables.

In steps 9 and 10, the average, minimum and maximum values are determined. If there are two or three variables, the block returns valid output values. If there is a valid value, the replacement value is returned.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	11,1
1510 SP F-1 PN	6,6
1511 F-1 PN / 1511 TF-1 PN	5,4
1515 F-2 PN / 1515 TF-2 PN	3,2
1516 F-3 PN/DP /1516 TF-3 PN/DP	2,1

4 Program blocks

CPU-Type	runtime [ms]
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,7
1518 F-4 PN/DP (T, ODK or MEP)	0,7

Bit	Description
0	Value State 1 invalid
1	Value State 2 invalid
2	Discrepancy time expired
3	Discrepancy fault
4	Acknowledge requested
5	First Cycle
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	Error Overflow, check internal status

operational sequence description

normal case

No.	Action	Comment
1	Set valueStates to true	
2	Set discrepancy time to 5 seconds and discrepancy 100	
3	Set input values to valid values in discrepancy	error is true, acknowledgeRequested is true, output is substituteValue
4	Set acknowledge to true	outputAverage, outputMax and outputMin is set

Change value in discrepancy time

Requirement: normal case, no faults

No.	Action	Comment
1	Set one input to an invalid value	outputAverage, outputMax and outputMin remain in valid values, error is false
2	Set the same input back to a valid input in the discrepancy time	outputAverage, outputMax and outputMin are set correctly, error is false. No acknowledge necessary
3	Set one input to an invalid value	outputAverage, outputMax and outputMin remain in valid values, error is false
4	Set the same input back to a valid input later than the discrepancy time	outputAverage, outputMax and outputMin are set to substituteValue1, error is true, acknowledgeRequested is true.

4 Program blocks

No.	Action	Comment
5	Set acknowledge to true	outputAverage, outputMax and outputMin are set correctly, error is false

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First

4.1.2 LSafe_2oo3AIDInt (SafetyFB / V1.0.0)

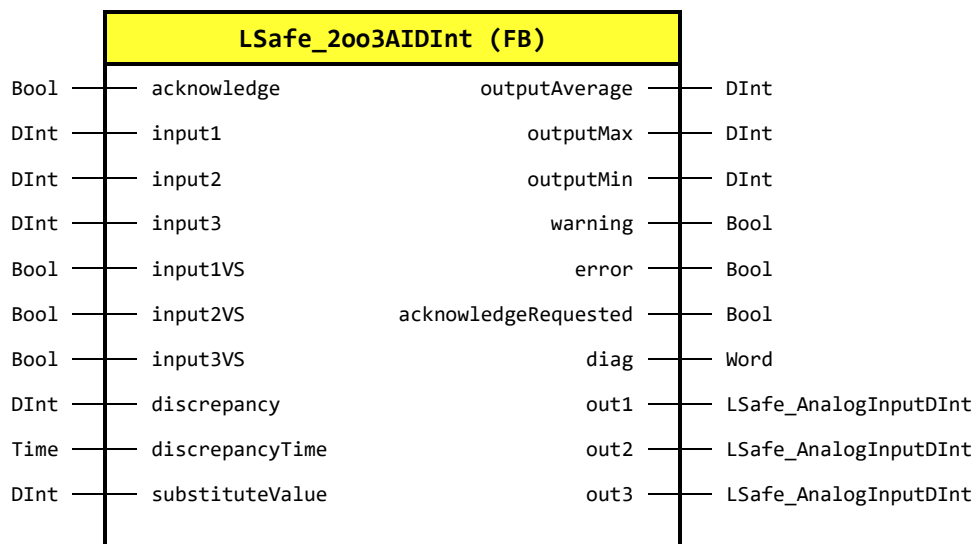
Author: Siemens

Short description

The block determines the arithmetic mean, the minimum and maximum value from three analog DINT input values. The arithmetic mean may only be used if all three values are valid, otherwise it must be checked for minimum and maximum values.

For three valid values, the arithmetic mean of all three is determined. The minimum and maximum values are respectively the smallest and largest of the three values. For two valid values, the arithmetic mean is determined from the two valid values. The minimum and maximum values are respectively the smallest and largest of the two values. For less than two valid values, the substitute value is output for the arithmetic mean, the minimum and maximum values. The block monitors whether the analog input values within the specified discrepancy time are within the specified discrepancy to each other. If this is not the case for an input value, it is deactivated until it is acknowledged. The block is acknowledged via the input acknowledge, which expects a positive edge. The acknowledgeRequested output informs accordingly about a block that can be acknowledged and, if necessary, reintegrated. The block can be acknowledged in the following three cases:

1. If the block can be restored to its functional state by acknowledging. Either by a valid variable and at least one variable to be acknowledged or by at least two acknowledgeable (re-integrable) variables.
2. If the block outputs a valid value (output) for two valid input values, but the third variable can also be acknowledged (reintegrated).
3. In the first cycle, three valid input values are required to reintegrate the device.

Interface description**Block interface****Input parameter**

Identifier	Data type	Default value	Description
acknowledge	Bool	FALSE	Acknowledgment input, positive signal required, NO.
input1	DInt	0	Analog input 1, (0 .. 700,000,000).

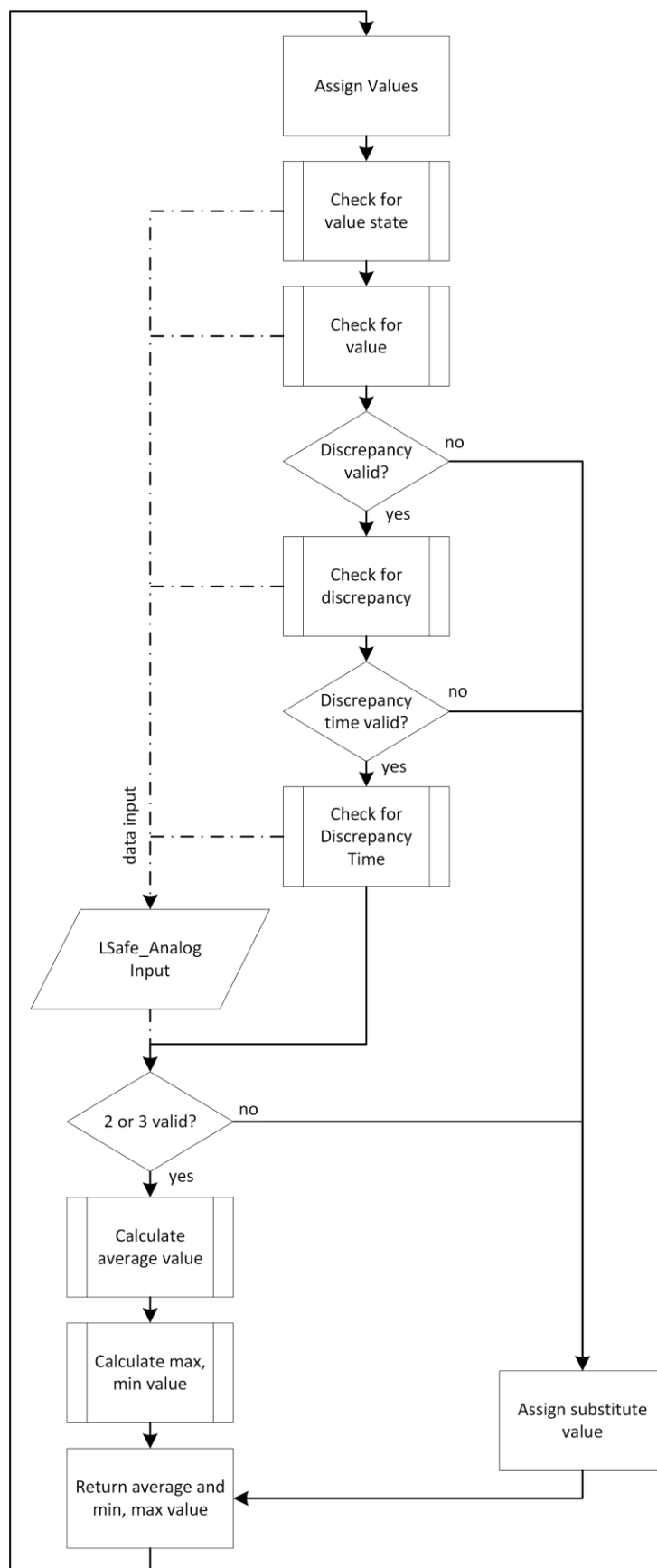
4 Program blocks

Identifier	Data type	Default value	Description
input2	DInt	0	Analog input 2, (0 .. 700,000,000).
input3	DInt	0	Analog input 3, (0 .. 700,000,000).
input1VS	Bool	FALSE	Value State of Input Value 1, False: fault.
input2VS	Bool	FALSE	Value State of Input Value 2, False: fault.
input3VS	Bool	FALSE	Value State of Input Value 3, False: fault.
discrepancy	DInt	0	Maximum distance between two analog values, the delta between all three analog values may be greater. The discrepancy can be a positive value between MIN_DISCREPANCY (0) and MAXIMUM_DISCREPANCY (700,000,000). Negative deltas result in the output of a surrogate value. Example: in1: 100; in2: 150; in3: 200; Delta: 60 would be valid, since two values are each less than or equal to 60 apart.
discrepancyTime	Time	T#0s	Time in which the values must be within the discrepancy. Time can range from 0ms to MAXIMUM_DISCREPANCY_TIME (60s by default). If one value is outside the discrepancy after the discrepancy time, a warning is issued, and two are caused by an error. An error in the discrepancy time must be acknowledged.
substituteValue	DInt	0	Replacement value. Output if less than two values are valid.

Output parameter

Identifier	Data type	Description
outputAverage	DInt	Arithmetic mean for two or three valid input values. If there are less than two valid values, the substitute value is output.
outputMax	DInt	If there are three valid variables, output of the maximum value. If there are two valid variables, output of the larger valid value. If there are less than two valid values, the substitute value is output.
outputMin	DInt	If there are three valid variables, the minimum value is output. If there are two valid variables, output of the smaller valid value. If there are less than two valid values, the substitute value is output.
warning	Bool	True, if at least one value property is invalid.
error	Bool	True, if more than one value property is invalid.
acknowledgeRequested	Bool	True, if at least one value property can be acknowledged. Acknowledging do not lead implicitly to valid outputs.
diag	Word	DIAG output.
out1	LSafe_AnalogInputDInt	Output of Input 1.
out2	LSafe_AnalogInputDInt	Output of Input 2.
out3	LSafe_AnalogInputDInt	Output of Input 3.

Functional description



NOTICE

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE

The enable input (EN) must not be wired for F-modules.

**WARNING**

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger.
Please note this if you want to change the settings shown here due to individual specifications.

Internal block description

In the first step, the block assigns substitute values to the internal variables and the external values.

In networks 2 – 7, the block checks various parameters of the analog input values. Furthermore, the attached discrepancy and discrepancy time are checked for validity.

In the second step, the block checks the value statuses and sets the input values to invalid if the value status is invalid.

In the third step, the input variables are checked for validity. They are checked for a range between or equal to MIN_VALUE and MAX_VALUE. The corresponding values are stored in the data type.

In the fourth step, the block checks the discrepancy for validity. If the input is invalid, the outputs are set to the equivalent values. The discrepancy may be between MIN_DISCREPANCY and MAX_DISCREPANCY.

In the fifth step, the block checks the variables for discrepancy. In this case, two variables may be separated at most in the specified discrepancy. The results are stored in the data type.

In the sixth step, the specified discrepancy time is checked for a maximum of MAX_DISCREPANCY_TIME (60s). If the value is invalid, the procedure is equivalent to the discrepancies that are invalid.

In the seventh step, timers are started when there is a discrepancy. If the input variables are not within the discrepancy within the discrepancy time, they are set to invalid and this information is cached in the data type.

In step 8 of the block, the input properties are checked for validity and correspond to the number of valid variables.

In steps 9 and 10, the average, minimum and maximum values are determined. If there are two or three variables, the block returns valid output values. If there is a valid value, the replacement value is returned.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	16,4
1510 SP F-1 PN	9,8
1511 F-1 PN / 1511 TF-1 PN	8,6
1515 F-2 PN / 1515 TF-2 PN	4,6
1516 F-3 PN/DP	3,2

4 Program blocks

CPU-Type	runtime [ms]
1517 F-3 PN/DP	1,0
1518 F-4 PN/DP (T, ODK or MEP)	1,0

Bit	Description
0	Value State 1 invalid
1	Value State 2 invalid
2	Value State 3 invalid
3	Discrepancy time expired
4	Discrepancy fault
5	Acknowledge requested
6	First Cycle
7	
8	
9	
10	
11	
12	
13	
14	
15	Error Overflow, check internal status

Operational sequence description

normal case

No.	Action	Comment
1	Set valueStates to true	
2	Set discrepancy time to 5 seconds and discrepancy 100	
3	Set input values to valid values in discrepancy	error is true, acknowledgeRequested is true, output is substituteValue
4	Set acknowledge to true	outputAverage, outputMax and outputMin is set

Change value in discrepancy time

Requirement: normal case, no faults

No.	Action	Comment
1	Set one input to an invalid value	outputAverage, outputMax and outputMin remain, warning is false
2	Set the same input back to a valid input in the discrepancy time	outputAverage, outputMax and outputMin are set according to all three valid values, warning is false. No acknowledge necessary
3	Set one input to an invalid value	outputAverage, outputMax and outputMin remain, error is false
4	Set the same input back to a valid input later than the discrepancy time	outputAverage, outputMax and outputMin are set to the two valid values, warning is true, acknowledgeRequested is true.

4 Program blocks

No.	Action	Comment
5	Set acknowledge to true	outputAverage, outputMax and outputMin are set correctly, error is false

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.1.3 LSafe_2oo3AIInt (SafetyFB / V1.0.0)

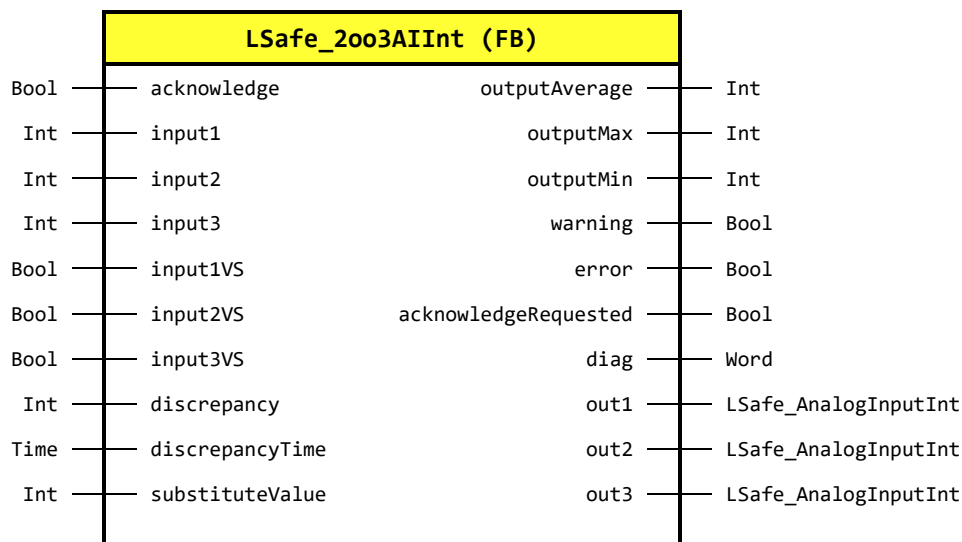
Author: Siemens

Short description

The block determines the arithmetic mean, the minimum and maximum value from three analog INT input values. The arithmetic mean may only be used if all three values are valid, otherwise it must be checked for minimum and maximum values.

For three valid values, the arithmetic mean of all three is determined. The minimum and maximum values are respectively the smallest and largest of the three values. For two valid values, the arithmetic mean is determined from the two valid values. The minimum and maximum values are respectively the smallest and largest of the two values. For less than two valid values, the substitute value is output for the arithmetic mean, the minimum and maximum values. The block monitors whether the analog input values within the specified discrepancy time are within the specified discrepancy to each other. If this is not the case for an input value, it is deactivated until it is acknowledged. The block is acknowledged via the input acknowledge, which expects a positive edge. The acknowledgeRequested output informs accordingly about a block that can be acknowledged and, if necessary, reintegrated. The block can be acknowledged in the following three cases:

1. If the block can be restored to its functional state by acknowledging. Either by a valid variable and at least one variable to be acknowledged or by at least two acknowledgeable (re-integrable) variables.
2. If the block outputs a valid value (output) for two valid input values, but the third variable can also be acknowledged (reintegrated).
3. In the first cycle, three valid input values are required to reintegrate the device.

Interface description**Block interface****Input parameter**

Identifier	Data type	Default value	Description
acknowledge	Bool	FALSE	Acknowledgment input, positive signal required, NO.
input1	Int	0	Analog input 1.

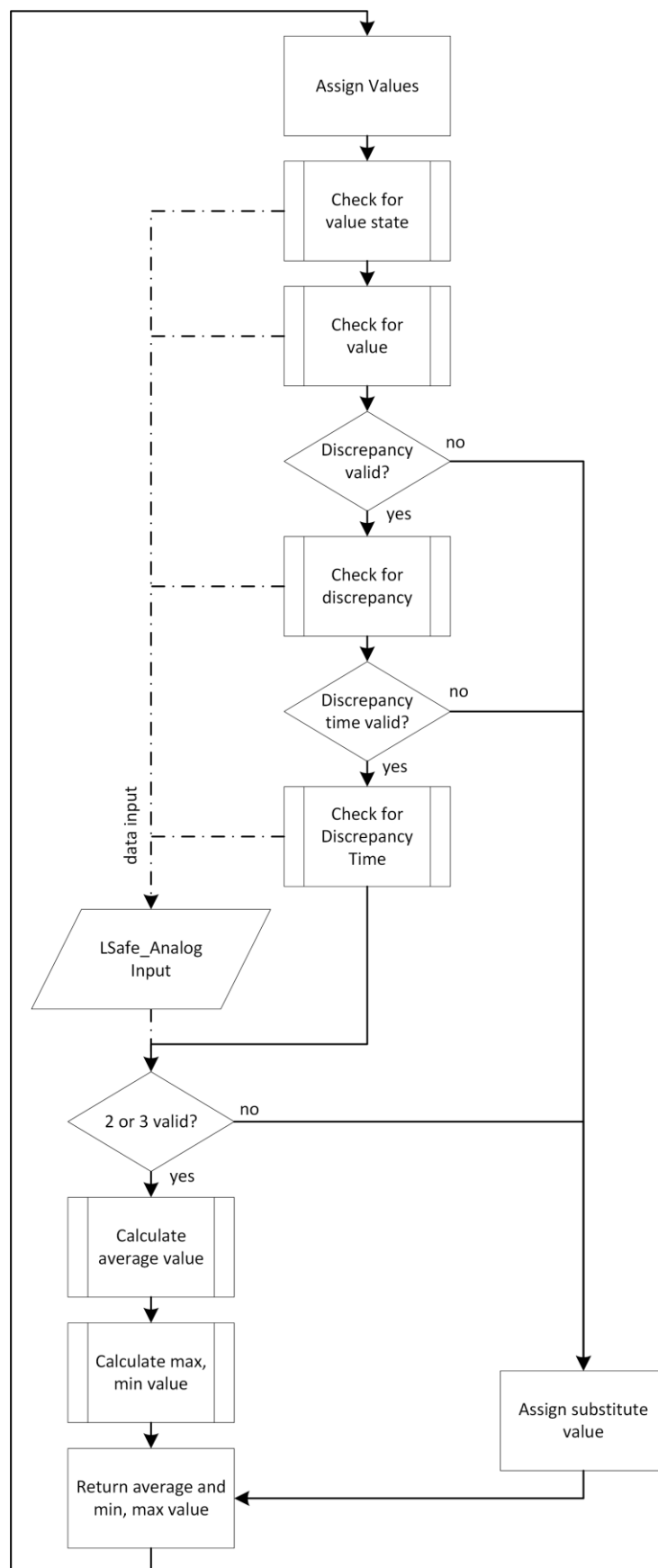
4 Program blocks

Identifier	Data type	Default value	Description
input2	Int	0	Analog input 2.
input3	Int	0	Analog input 3.
input1VS	Bool	FALSE	Value State of Input Value 1, False: fault.
input2VS	Bool	FALSE	Value State of Input Value 2, False: fault.
input3VS	Bool	FALSE	Value State of Input Value 3, False: fault.
discrepancy	Int	0	Maximum distance between two analog values, the delta between all three analog values may be greater. The discrepancy can be a positive value between MIN_DISCREPANCY (0) and MAXIMUM_DISCREPANCY (32,767). Negative deltas result in the output of a surrogate value. Example: in1: 100; in2: 150; in3: 200; Delta: 60 would be valid, since two values are each less than or equal to 60 apart.
discrepancyTime	Time	T#0s	Time in which the values must be within the discrepancy. Time can range from 0ms to MAXIMUM_DISCREPANCY_TIME (60s by default). If one value is outside the discrepancy after the discrepancy time, a warning is issued, and two are caused by an error. An error in the discrepancy time must be acknowledged.
substituteValue	Int	0	Replacement value. Output if less than two values are valid.

Output parameter

Identifier	Data type	Description
outputAverage	Int	Output of arithmetic mean. Returns substitute value if less than two inputs are valid.
outputMax	Int	Outputs the maximum value of two or three valid inputs.
outputMin	Int	Outputs the minimum value of two or three valid inputs.
warning	Bool	True, if at least one value property is invalid.
error	Bool	True, if at least two value properties of at least two variables are invalid.
acknowledgeRequested	Bool	True, if at least one value property can be acknowledged. Acknowledging do not lead implicitly to valid outputs.
diag	Word	DIAG output.
out1	LSafe_AnalogInputInt	Output of Input 1.
out2	LSafe_AnalogInputInt	Output of Input 2.
out3	LSafe_AnalogInputInt	Output of Input 3.

Functional description



NOTICE

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE

The enable input (EN) must not be wired for F-modules.

**WARNING**

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger.
Please note this if you want to change the settings shown here due to individual specifications.

Internal block description

In the first step, the block assigns substitute values to the internal variables and the external values.

In networks 2 – 7, the block checks various parameters of the analog input values. Furthermore, the attached discrepancy and discrepancy time are checked for validity.

In the second step, the block checks the value statuses and sets the input values to invalid if the value status is invalid.

In the third step, the input variables are checked for validity. They are checked for a range between or equal to MIN_VALUE and MAX_VALUE. The corresponding values are stored in the data type.

In the fourth step, the block checks the discrepancy for validity. If the input is invalid, the outputs are set to the equivalent values. The discrepancy may be between MIN_DISCREPANCY and MAX_DISCREPANCY.

In the fifth step, the block checks the variables for discrepancy. In this case, two variables may be separated at most in the specified discrepancy. The results are stored in the data type.

In the sixth step, the specified discrepancy time is checked for a maximum of MAX_DISCREPANCY_TIME (60s). If the value is invalid, the procedure is equivalent to the discrepancies that are invalid.

In the seventh step, timers are started when there is a discrepancy. If the input variables are not within the discrepancy within the discrepancy time, they are set to invalid and this information is cached in the data type.

In step 8 of the block, the input properties are checked for validity and correspond to the number of valid variables.

In steps 9 and 10, the average, minimum and maximum values are determined. If there are two or three variables, the block returns valid output values. If there is a valid value, the replacement value is returned.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	16,4
1510 SP F-1 PN	9,7
1511 F-1 PN / 1511 TF-1 PN	8,6
1515 F-2 PN / 1515 TF-2 PN	4,6
1516 F-3 PN/DP / 1516 F-3 PN/DP	3,1

4 Program blocks

CPU-Type	runtime [ms]
1517 F-3 PN/DP / 1517 F-3 PN/DP	1,0
1518 F-4 PN/DP (T, ODK or MEP)	1,0

Bit	Description
0	Value State 1 invalid
1	Value State 2 invalid
2	Value State 3 invalid
3	Discrepancy time expired
4	Discrepancy fault
5	Acknowledge requested
6	First Cycle
7	
8	
9	
10	
11	
12	
13	
14	
15	Error Overflow, check internal status

Operational sequence description

normal case

No.	Action	Comment
1	Set valueStates to true	
2	Set discrepancy time to 5 seconds and discrepancy 100	
3	Set input values to valid values in discrepancy	error is true, acknowledgeRequested is true, output is substituteValue
4	Set acknowledge to true	outputAverage, outputMax and outputMin is set

Change value in discrepancy time

Requirement: normal case, no faults

No.	Action	Comment
1	Set one input to an invalid value	outputAverage, outputMax and outputMin remain, warning is false
2	Set the same input back to a valid input in the discrepancy time	outputAverage, outputMax and outputMin are set according to all three valid values, warning is false. No acknowledge necessary
3	Set one input to an invalid value	outputAverage, outputMax and outputMin remain, error is false
4	Set the same input back to a valid input later than the discrepancy time	outputAverage, outputMax and outputMin are set to the two valid values, warning is true, acknowledgeRequested is true.

4 Program blocks

No.	Action	Comment
5	Set acknowledge to true	outputAverage, outputMax and outputMin are set correctly, error is false

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First

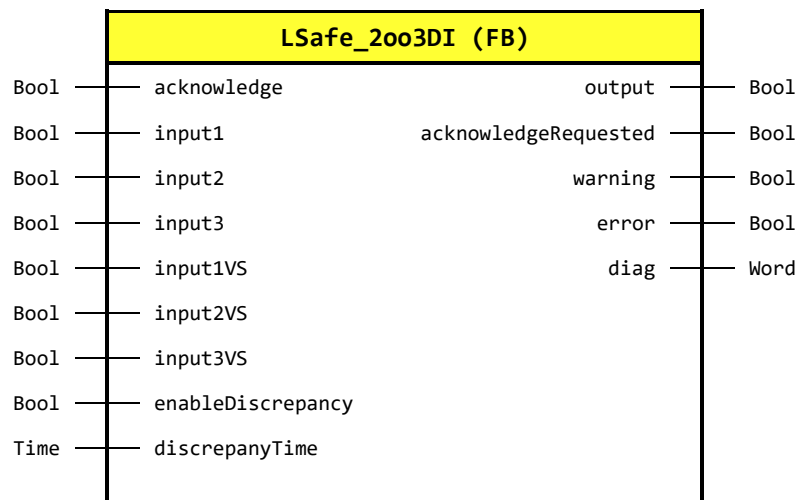
4.1.4 LSafe_2oo3DI (SafetyFB / V1.0.0)

Short description

The function block determines an output value from three digital input values. For less than two valid input values, the function block returns the substitute value; for two or three valid input values, the function block returns the output that corresponds to the majority of the input values. If the discrepancy time is activated, at least two values must receive the same value within the valid time, otherwise an error is output. To leave the first cycle, all three inputs needs to be valid and true.

Interface description

Block interface



Input parameter

Identifier	Data type	Default value	Description
acknowledge	Bool	FALSE	Acknowledge, positive signal edge, NO.
input1	Bool	FALSE	Input 1 Bool, NO.
input2	Bool	FALSE	Input 2 Bool, NO.
input3	Bool	FALSE	Input 3 Bool, NO.
input1VS	Bool	FALSE	Value state Input 1, NC, true: value correct.
input2VS	Bool	FALSE	Value state Input 2, NC, true: value correct.
input3VS	Bool	FALSE	Value state Input 3, NC, true: value correct.
enableDiscrepancy	Bool	FALSE	Enable Discrepancy Test, NO.
discrepanyTime	Time	T#0MS	Discrepancy time for all three input variables in ms.

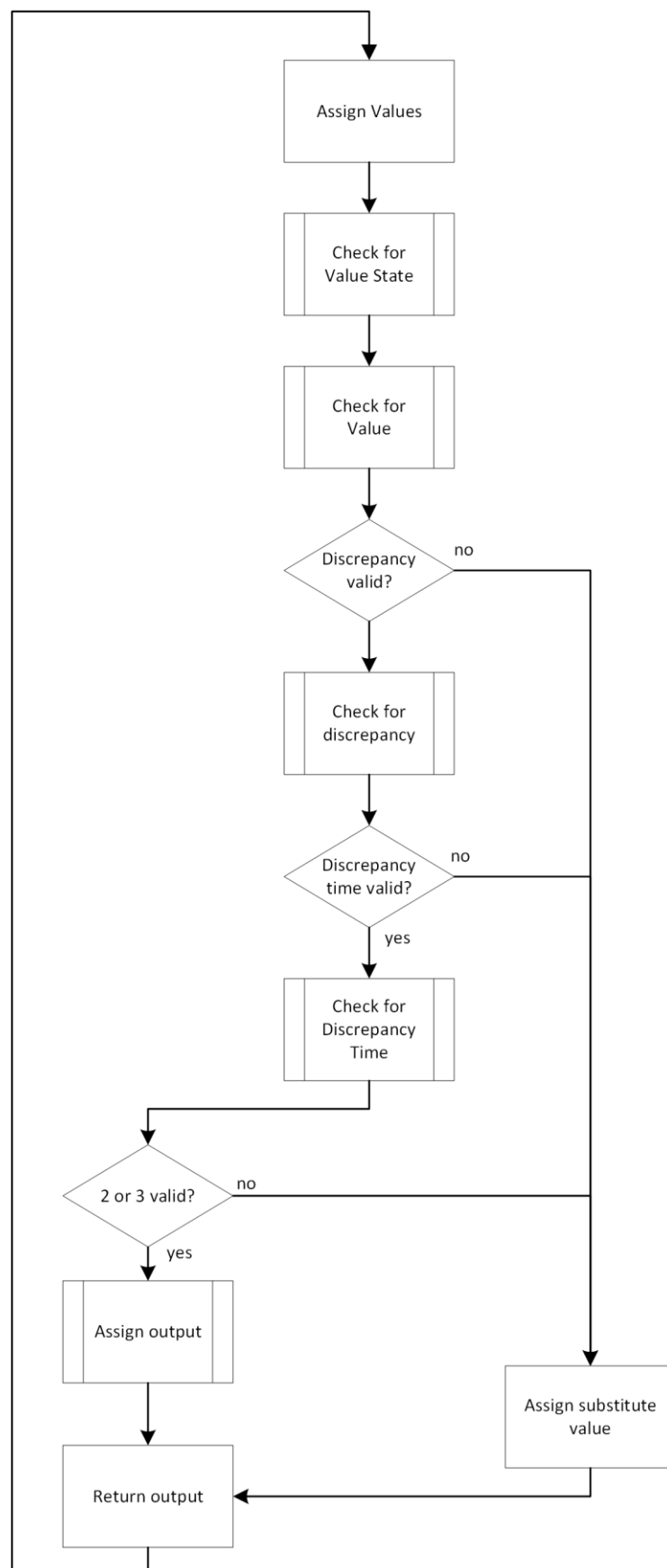
Output parameter

Identifier	Data type	Description
output	Bool	Actuator output, 2oo3.
acknowledgeRequested	Bool	True, if acknowledge requested after reintegration.
warning	Bool	True, if at least one value propertie is invalid.

4 Program blocks

Identifier	Data type	Description
error	Bool	True, if at least two value properties of at least two values are invalid.
diag	Word	DIAG output for non-safety diagnostic information.

Functional description



NOTICE To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE The enable input (EN) must not be wired for F-modules.



WARNING

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Procedure description

In the first step, the outputs are pre-assigned with default values.

In the second step, the function block checks the value status and sets the input values to invalid if the value status is invalid.

In the third step, the input variables are checked for validity.

The fourth step checks whether the discrepancy time entered is valid, regardless of whether the inputs are to be checked for discrepancy time or not.

If the discrepancy time is to be checked, this check is carried out immediately afterwards.

In the fifth step, the function block checks how many inputs are valid. If 2 or 3 inputs are valid, they are evaluated further, otherwise substitute values are assigned.

In the sixth step, the majority of valid inputs are output. If fewer than two values are valid, the substitute value is assigned accordingly.

In the seventh and final step, the assigned values are written to the function block output.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	3,4
1510 SP F-1 PN	1,7
1511 F-1 PN / 1511 TF-1 PN	1,4
1515 F-2 PN / 1515 TF-2 PN	0,8
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,4
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK or MEP)	0,1

Bit	Description
0	Value State 1 invalid
1	Value State 2 invalid
2	Value State 3 invalid
3	Discrepancy 1 invalid
4	Discrepancy 2 invalid
5	Discrepancy 3 invalid

4 Program blocks

Bit	Description
6	Discrepancy Time Invalid 1
7	Discrepancy Time Invalid 2
8	Discrepancy Time Invalid 3
9	Discrepancy Time Invalid for all inputs
10	
11	
12	
13	
14	
15	Acknowledge requested

operational sequence description

normal case

No.	Action	Comment
1	Set valueStates to true	
2	Set all three inputs to true	Acknowledge requested is true
3	Set acknoweldge to true	output is set

Change value in discrepancy time

Requirement: normal case, no faults; enableDiscrepancy is true, discrepancyTime is greater than 0

No.	Action	Comment
1	Set one input to an invalid value	output is set according the valid values, warning to true
2	Set the same input back to a valid input in the discrepancy time	output is set according to all three valid values, warning is false. No acknoweldge necessary
3	Set one input to an invalid value	output is set to substituteValue1, error to true
4	Set the same input back to a valid input later than the discrepancy time	output is set to the two valid values, warning is true, acknoweldgeRequested is true.
5	Set acknowledge to true	output is correctly, error is false

Change log

Version & Date	Change description
01.00.00	Siemens Industry Online Support
31.02.2024	First released

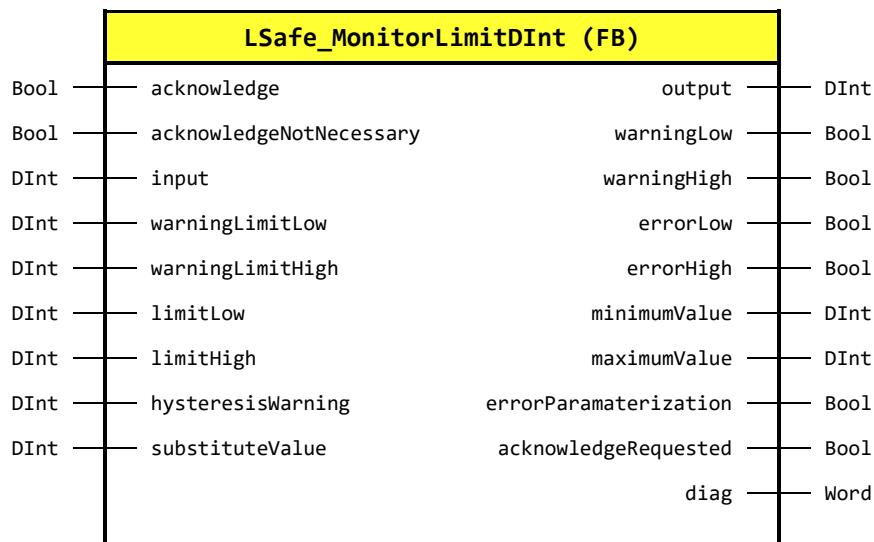
4.1.5 LSafe_MonitorLimitDInt (SafetyFB / V1.0.0)

Short description

The block evaluates the double integer input regarding to the warning-limits and limits. A distinction is made between warning-limits and limits. The limits must be acknowledged if acknowledgeNotNecessary is set to false. This leads to an output of the errors: "errorLow" or "errorHigh". If one of the errors is triggered, "substituteValue" is written to the output. The warning-limits lead to the output of the warning outputs: "warningLow" or "warningHigh" if they are exceeded or not reached. These do not have to be acknowledged. Optionally, a hysteresis can be applied to the warning limits. If activated, the warnings are only reset if the input value is in the good range with a distance of the hysteresis, i.e. higher than "warningLow" plus "hysteresisWarning" or lower than "warningLow" minus "hysteresisWarning". If no hysteresis should be used, the value can be set to 0. The "errorParamaterization" output is set to true if the lower warning-limits or limits are greater than the upper ones or a negative hysteresis is paramaterized. To exit the first cycle, the input value must be within the warning limits if acknowledgement is necessary (acknowledgeNotNecessary = false).

Interface description

Block interface



Input parameter

Identifier	Data type	Default value	Description
acknowledge	Bool	FALSE	Acknowledge, positive edge signal, NO.
acknowledgeNotNecessary	Bool	FALSE	True, if acknowledge for faults is not necessary.
input	DInt	0	Input value.
warningLimitLow	DInt	0	Warning limit, low value.
warningLimitHigh	DInt	0	Warning limit, high value.
limitLow	DInt	0	Limit, low value.
limitHigh	DInt	0	Limit, high value.

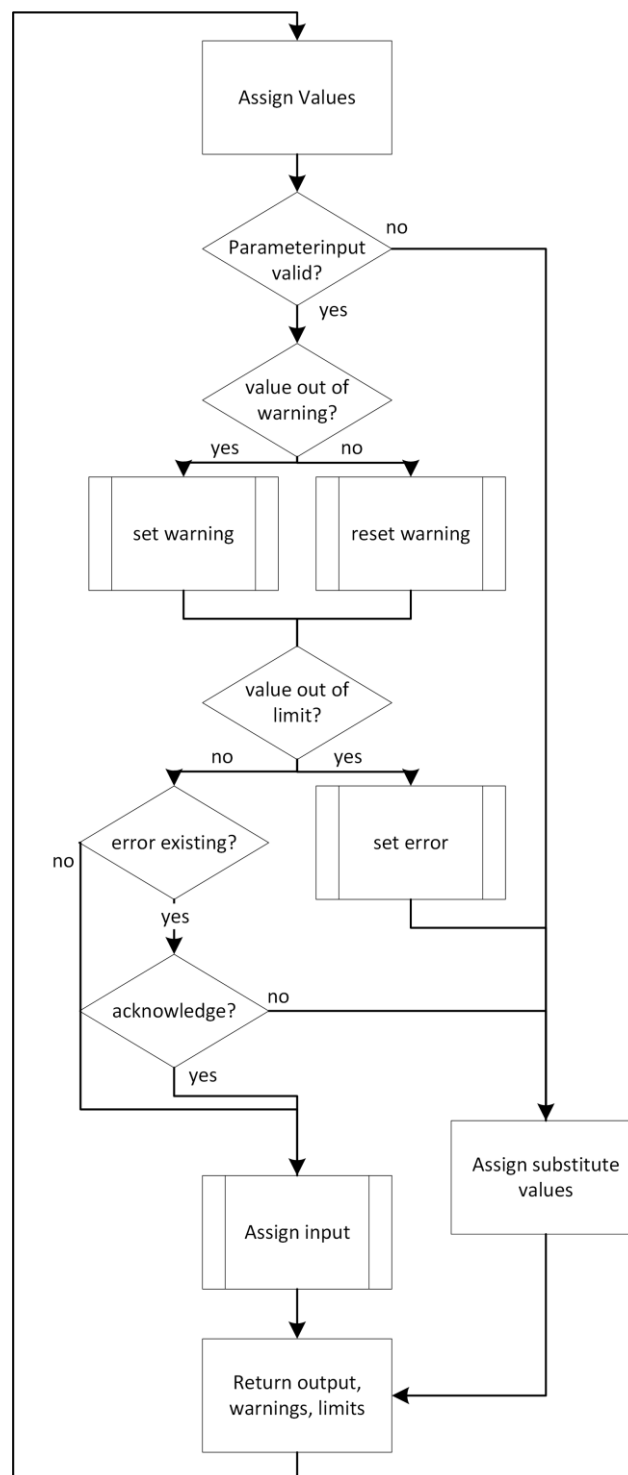
4 Program blocks

Identifier	Data type	Default value	Description
hysteresisWarning	DInt	0	Hysteresis for Warning value, warningLimitLow plus hysteresis or warningLimitHigh minus hysteresis will end the warning output.
substituteValue	DInt	0	Substitute value if error.

Output parameter

Identifier	Data type	Description
output	DInt	Output value, input if normal, substitute value if error.
warningLow	Bool	True, when input lower than warningLimitLow.
warningHigh	Bool	True, when input higher than warningLimitHigh.
errorLow	Bool	True, when input lower than limitLow.
errorHigh	Bool	True, when input higher than limitHigh.
minimumValue	DInt	Minimum value, when error. Lowest value remains until acknowledge of errorLow.
maximumValue	DInt	Maximum value, when error. Highest value remains until acknowledge of errorHigh.
errorParameterization	Bool	True, when internal fault for false input warning, limits or discrepancy.
acknowledgeRequested	Bool	True, when block can be acknowledged.
diag	Word	Diagnose output for status information.

Functional description

**NOTICE**

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE

The enable input (EN) must not be wired for F-modules.

**WARNING**

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger.
Please note this if you want to change the settings shown here due to individual specifications.

Internal block description

In the first step, the outputs are pre-assigned with default values.

In the second step, the parameterization of the module is checked. The inputs for limits, limits and hysteresis are included for this purpose.

In the third step, the input value is checked for the warning limits (warningLimitLow and warningLimitHigh). The warningLow and warningHigh outputs are pre-assigned according to the result.

In the fourth step, the input value is checked for the limit limits (limitLow and limitHigh) and the outputs errorLow and errorHigh are pre-assigned according to the results.

In the fifth step, as long as all conditions are met, earlier errors can be acknowledged.

In the sixth and final step, the assigned values are written to the function block output.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	13,2
1510 SP F-1 PN	7,5
1511 F-1 PN / 1511 TF-1 PN	6,6
1515 F-2 PN / 1515 TF-2 PN	3,8
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK or MEP)	0,9

Bit	Description
0	Parameterization error maximum value
1	Parameterization error minimum value
2	Parameterization error warning values
3	Parameterization error hysteresis lower than zero
4	Parameterization error substitute value
5	Initialization/ First cycle
6	
7	
8	
9	
10	
11	
12	
13	

4 Program blocks

Bit	Description
14	
15	Error Overflow, check internal status

operational sequence description

normal case

No.	Action	Comment
1	choose warningLimitLow = 250, warningLimitHigh = 750, limitLow = 100, limitHigh = 900, input = 500	acknowledgeRequested is true, output is substituteValue, warnings and errors are true
2	set acknowledge to true	output is input, warnings and errors are false

Get into warning limit low

Requirement: normal case, input is in valid, values as in normal case, no faults, hysteresisWarning is 0

No.	Action	Comment
1	set input to 200	output is 200, warningLow is true
2	reset input to 500	output is 500, warningLow is false

Get into warning limit low with activated hysteresisWarning

Requirement: normal case, input is in valid, values as in normal case, no faults, hysteresisWarning is 300

No.	Action	Comment
1	set input to 200	output is 200, warningLow is true
2	reset input to 500	output is 500, warningLow is true
3	set input to 600	output is 600, warningLow is false

Get into error low

Requirement: normal case, input is in valid, values as in normal case, no faults

No.	Action	Comment
1	set input to 50	output is substituteValue, warningLow and errorLow are true, minimumValue is set to 50
2	reset input to 500	output is substituteValue, warningLow is false, errorLow is true, acknowledgeRequested is true
3	set acknowledge	output is 500, errorLow is false, minimumValue is 1 (default value)

Get into error low with acknowledgeNotNecessary activated

Requirement: normal case, input is in valid, values as in normal case, no faults

No.	Action	Comment
1	set input to 50	output is substituteValue, warningLow and errorLow are true, minimumValue is set to 50
2	reset input to 500	output is 500, errorLow is false, minimumValue is 1 (default value)

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

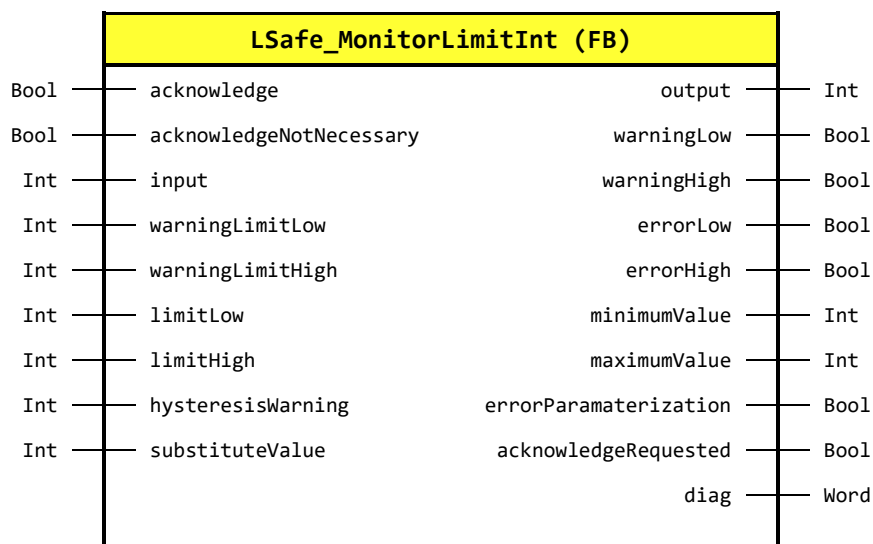
4.1.6 LSafe_MonitorLimitInt (SafetyFB / V1.0.0)

Short description

The block evaluates the integer input regarding to the warning-limits and limits. A distinction is made between warning-limits and limits. The limits must be acknowledged if acknowledgeNotNecessary is set to false. This leads to an output of the errors: "errorLow" or "errorHigh". If one of the errors is triggered, "substituteValue" is written to the output. The warning-limits lead to the output of the warning outputs: "warningLow" or "warningHigh" if they are exceeded or not reached. These do not have to be acknowledged. Optionally, a hysteresis can be applied to the warning limits. If activated, the warnings are only reset if the input value is in the good range with a distance of the hysteresis, i.e. higher than "warningLow" plus "hysteresisWarning" or lower than "warningLow" minus "hysteresisWarning". If no hysteresis should be used, the value can be set to 0. The "errorParamaterization" output is set to true if the lower warning-limits or limits are greater than the upper ones or a negative hysteresis is parameterized. To exit the first cycle, the input value must be within the warning limits if acknowledgement is necessary (acknowledgeNotNecessary = false).

Interface description

Block interface



Input parameter

Identifier	Data type	Default value	Description
acknowledge	Bool	FALSE	Acknowledge, positive edge signal, NO.
acknowledgeNotNecessary	Bool	FALSE	True, if acknowledge for faults is not necessary.
input	Int	0	Input value.
warningLimitLow	Int	0	Warning limit, low value.
warningLimitHigh	Int	0	Warning limit, high value.
limitLow	Int	0	Limit, low value.
limitHigh	Int	0	Limit, high value.

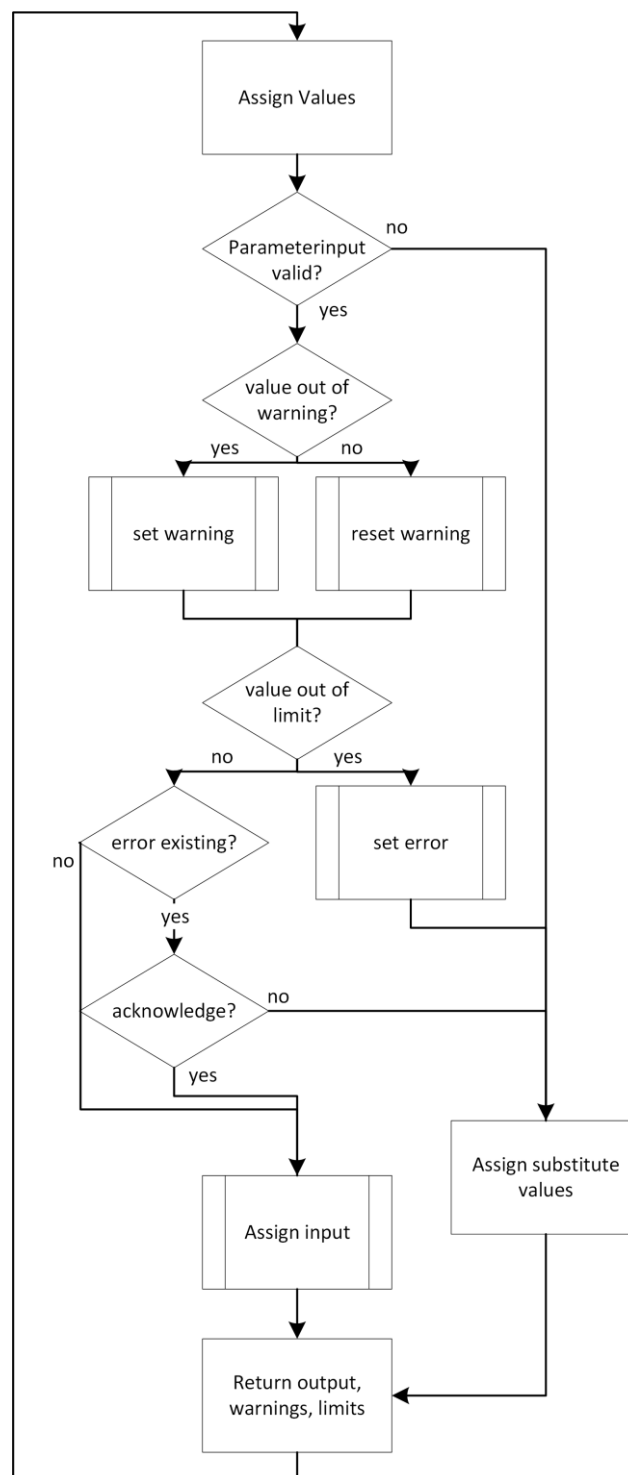
4 Program blocks

Identifier	Data type	Default value	Description
hysteresisWarning	Int	0	Hysteresis for Warning value, warningLimitLow plus hysteresis oder warningLimitHigh minus hysteresis will end the warning output.
substituteValue	Int	0	Substitute value if error. Must be outside of limits.

Output parameter

Identifier	Data type	Description
output	Int	Output value, input if normal, substitute value if error.
warningLow	Bool	True, when input lower than warningLimitLow.
warningHigh	Bool	True, when input higher than warningLimitHigh.
errorLow	Bool	True, when input lower than limitLow.
errorHigh	Bool	True, when input higher than limitHigh.
minimumValue	Int	Minimum value, when error. Lowest value remains until acknowledge of errorLow.
maximumValue	Int	Maximum value, when error. Highest value remains until acknowledge of errorHigh.
errorParamaterization	Bool	True, when internal fault for false input warning, limits or discrepancy.
acknowledgeRequested	Bool	True, when block can be acknowledged.
diag	Word	Diagnose output for status information.

Functional description

**NOTICE**

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE

The enable input (EN) must not be wired for F-modules.

**WARNING**

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger.
Please note this if you want to change the settings shown here due to individual specifications.

Internal block description

In the first step, the outputs are pre-assigned with default values.

In the second step, the parameterization of the module is checked. The inputs for limits, limits and hysteresis are included for this purpose.

In the third step, the input value is checked for the warning limits (warningLimitLow and warningLimitHigh). The warningLow and warningHigh outputs are pre-assigned according to the result.

In the fourth step, the input value is checked for the limit limits (limitLow and limitHigh) and the outputs errorLow and errorHigh are pre-assigned according to the results.

In the fifth step, as long as all conditions are met, earlier errors can be acknowledged.

In the sixth and final step, the assigned values are written to the function block output.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	13,2
1510 SP F-1 PN	7,5
1511 F-1 PN / 1511 TF-1 PN	6,6
1515 F-2 PN / 1515 TF-2 PN	3,8
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK or MEP)	0,9

Bit	Description
0	Parameterization error maximum value
1	Parameterization error minimum value
2	Parameterization error warning values
3	Parameterization error hysteresis lower than zero
4	Parameterization error substitute value
5	Initialization/ First cycle
6	
7	
8	
9	
10	
11	
12	
13	

4 Program blocks

Bit	Description
14	
15	Error Overflow, check internal status

operational sequence description

normal case

No.	Action	Comment
1	choose warningLimitLow = 250, warningLimitHigh = 750, limitLow = 100, limitHigh = 900, input = 500	acknowledgeRequested is true, output is substituteValue, warnings and errors are true
2	set acknowledge to true	output is input, warnings and errors are false

Get into warning limit low

Requirement: normal case, input is in valid, values as in normal case, no faults, hysteresisWarning is 0

No.	Action	Comment
1	set input to 200	output is 200, warningLow is true
2	reset input to 500	output is 500, warningLow is false

Get into warning limit low with activated hysteresisWarning

Requirement: normal case, input is in valid, values as in normal case, no faults, hysteresisWarning is 300

No.	Action	Comment
1	set input to 200	output is 200, warningLow is true
2	reset input to 500	output is 500, warningLow is true
3	set input to 600	output is 600, warningLow is false

Get into error low

Requirement: normal case, input is in valid, values as in normal case, no faults

No.	Action	Comment
1	set input to 50	output is substituteValue, warningLow and errorLow are true, minimumValue is set to 50
2	reset input to 500	output is substituteValue, warningLow is false, errorLow is true, acknowledgeRequested is true
3	set acknowledge	output is 500, errorLow is false, minimumValue is 1 (default value)

Get into error low with acknowledgeNotNecessary activated

Requirement: normal case, input is in valid, values as in normal case, no faults

No.	Action	Comment
1	set input to 50	output is substituteValue, warningLow and errorLow are true, minimumValue is set to 50
2	reset input to 500	output is 500, errorLow is false, minimumValue is 1 (default value)

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.2 Functions

4.2.1 LSafe_AverageOfThreeDInt (SafetyFC / V1.0.0)

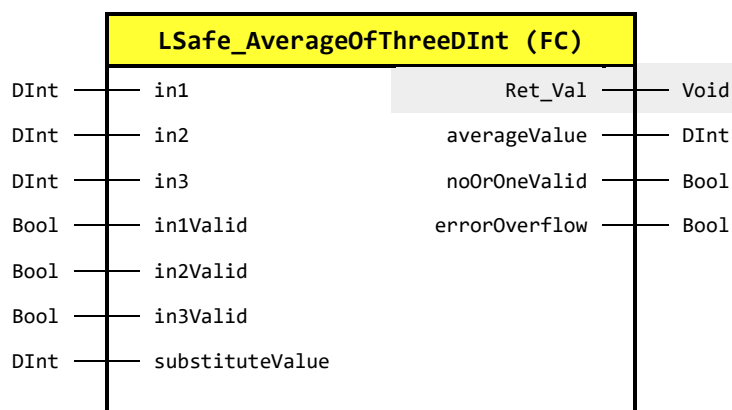
Author: Siemens

Short description

Returns the arithmetic average value from up to three valid inputs. Returns noOrOneValid true and the substitute value, if less than two input variables are valid.

Interface description

Block interface



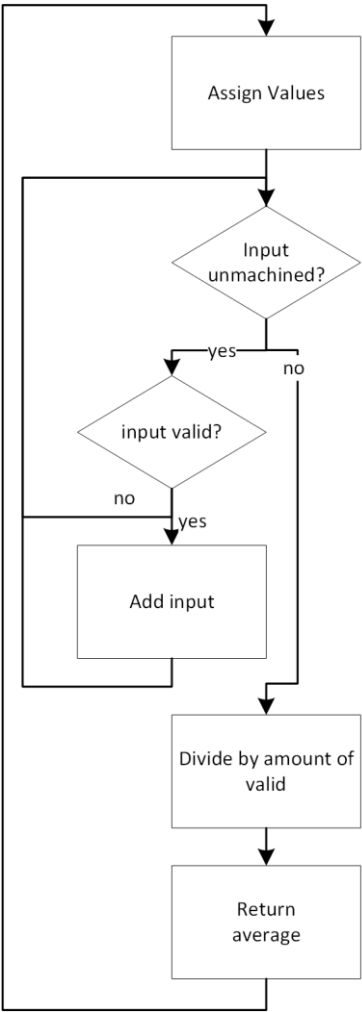
Input parameter

Identifier	Data type	Description
in1	DInt	Analog input 1.
in2	DInt	Analog input 2.
in3	DInt	Analog input 3.
in1Valid	Bool	True, if input 1 is valid.
in2Valid	Bool	True, if input 2 is valid.
in3Valid	Bool	True, if input 3 is valid.
substituteValue	DInt	Substitute value, will be returned, if less than two input values valid.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
averageValue	DInt	Average value of inputs. Will be the substitute value, if less than two inputs valid.
noOrOneValid	Bool	True, if less than two inputs are valid.
errorOverflow	Bool	True, if overflow error occurs.

Functional description



NOTICE To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE The enable input (EN) must not be wired for F-modules.

 **WARNING** Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	1,6
1510 SP F-1 PN	0,9
1511 F-1 PN / 1511 TF-1 PN	0,8

4 Program blocks

CPU-Type	runtime [ms]
1515 F-2 PN / 1515 TF-2 PN	0,5
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK or MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.2.2 LSafe_AverageOfThreeInt (SafetyFC / V1.0.0)

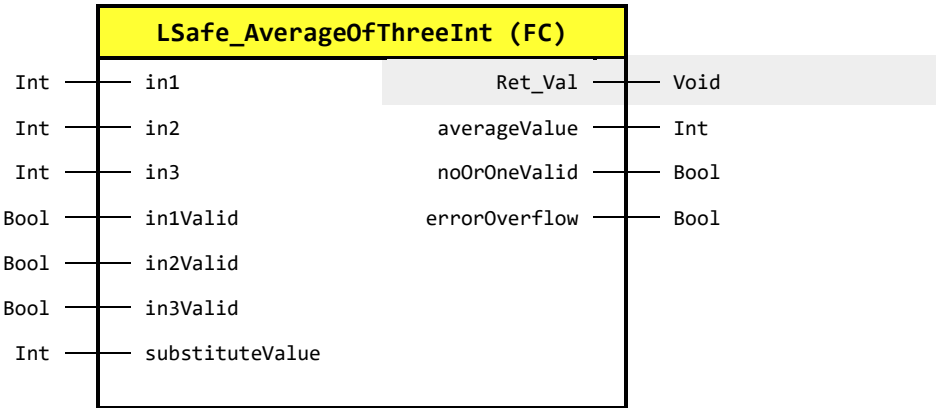
Author: Siemens

Short description

Returns the arithmetic average value from up to three valid inputs. Returns noOrOneValid true and the substitute value, if less than two input variables are valid.

Interface description

Block interface



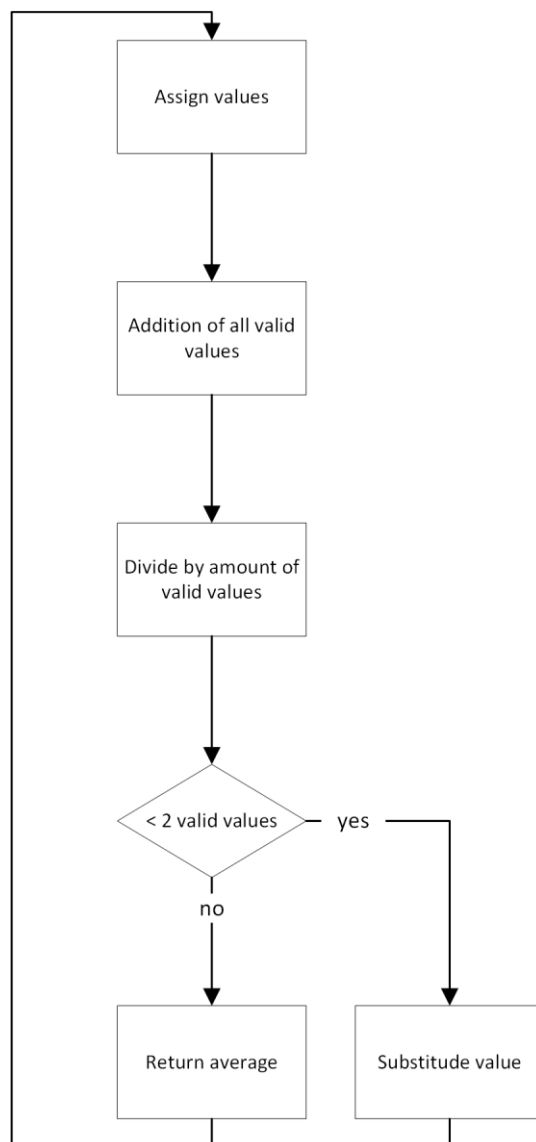
Input parameter

Identifier	Data type	Description
in1	Int	Analog input 1.
in2	Int	Analog input 2.
in3	Int	Analog input 3.
in1Valid	Bool	True, if input 1 is valid.
in2Valid	Bool	True, if input 2 is valid.
in3Valid	Bool	True, if input 3 is valid.
substituteValue	Int	Substitutue value, will be returned, if less than two input values valid.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
averageValue	Int	Average value of inputs. Will be the substitute value, if less than two inputs valid.
noOrOneValid	Bool	True, if less than two inputs are valid.
errorOverflow	Bool	True, if overflow error occurs.

Functional description



NOTICE

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE

The enable input (EN) must not be wired for F-modules.



WARNING

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	1,6
1510 SP F-1 PN	0,9
1511 F-1 PN / 1511 TF-1 PN	0,8
1515 F-2 PN / 1515 TF-2 PN	0,5
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.2.3 LSafe_InRangeDInt (SafetyFC / V1.0.0)

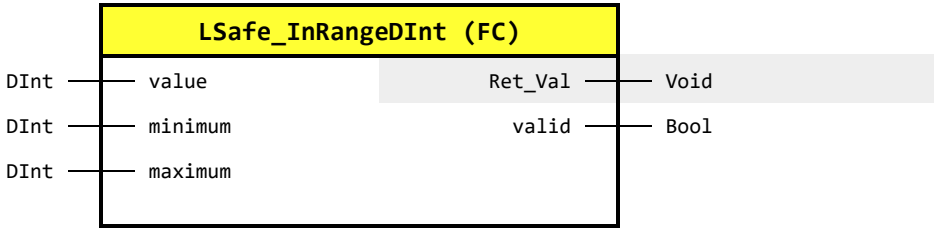
Author: Siemens

Short description

Evaluates if value is in between minimum and maximum and returns true if so.

Interface description

Block interface



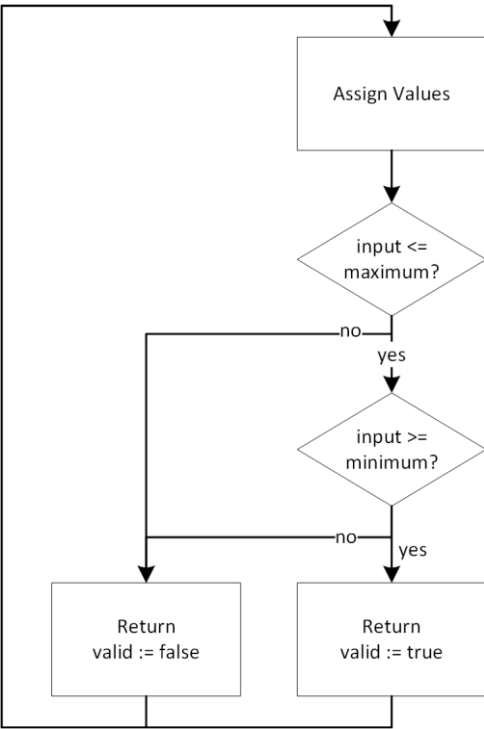
Input parameter

Identifier	Data type	Description
value	DInt	Input value.
minimum	DInt	Input minimum value, value has to be greater than or equal as minimum.
maximum	DInt	Input maximum value, value has to be sameller than or equal as maximum.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
valid	Bool	Output bit, true if value in range.

Functional description



NOTICE To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE The enable input (EN) must not be wired for F-modules.

 **WARNING** Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	1,4
1510 SP F-1 PN	0,9
1511 F-1 PN / 1511 TF-1 PN	0,8
1515 F-2 PN / 1515 TF-2 PN	0,4
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.2.4 LSafe_InRangeInt (SafetyFC / V1.0.0)

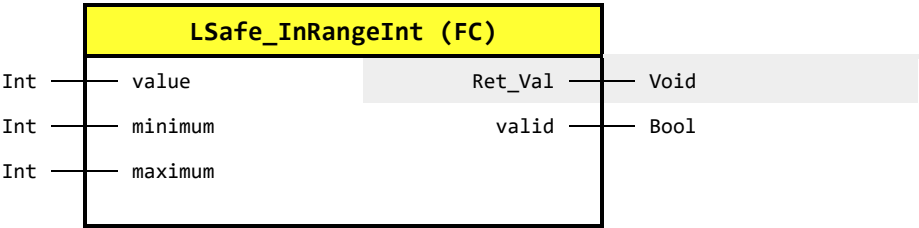
Author: Siemens

Short description

Evaluates if value is in between minimum and maximum.

Interface description

Block interface



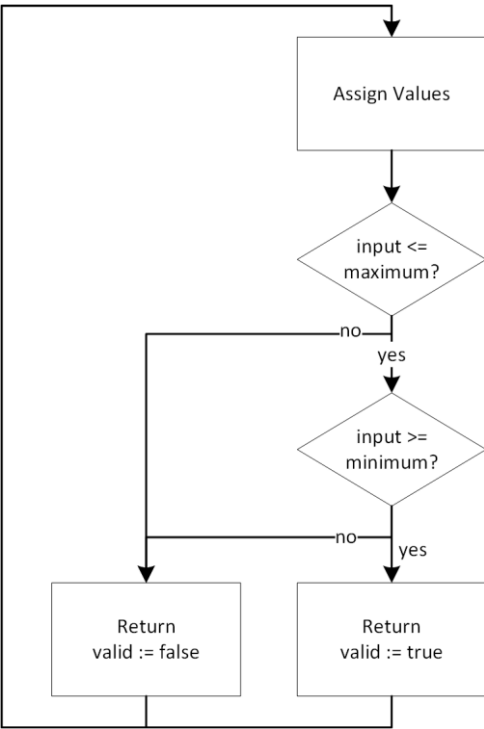
Input parameter

Identifier	Data type	Description
value	Int	Input value.
minimum	Int	Input minimum value, value has to be greater than or equal as minimum.
maximum	Int	Input maximum value, value has to be smaller than or equal as maximum.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
valid	Bool	Output bit, true if value in range.

Functional description



NOTICE To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE The enable input (EN) must not be wired for F-modules.

 **WARNING** Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	1,4
1510 SP F-1 PN	0,9
1511 F-1 PN / 1511 TF-1 PN	0,8
1515 F-2 PN / 1515 TF-2 PN	0,4
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.2.5 LSafe_MaxMinOfThreeDInt (SafetyFC / V1.0.0)

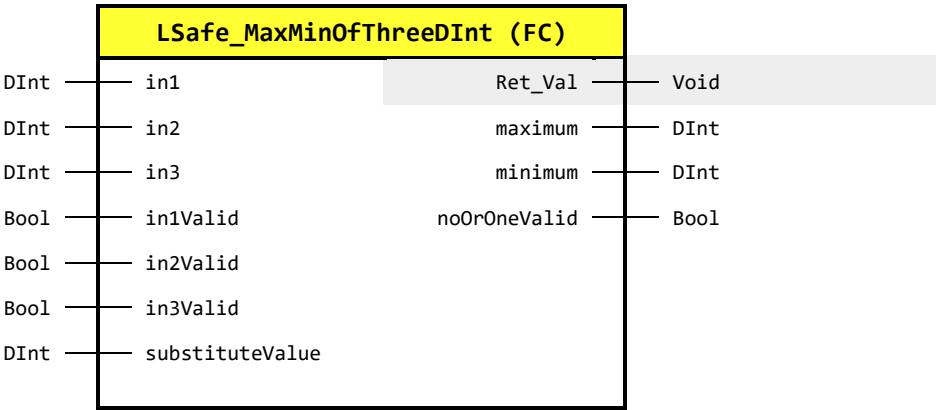
Author: Siemens

Short description

The function block returns the maximum and minimum value for two or three valid DoubleInteger variables. If less than two variables are valid, the substitute value is returned.

Interface description

Block interface



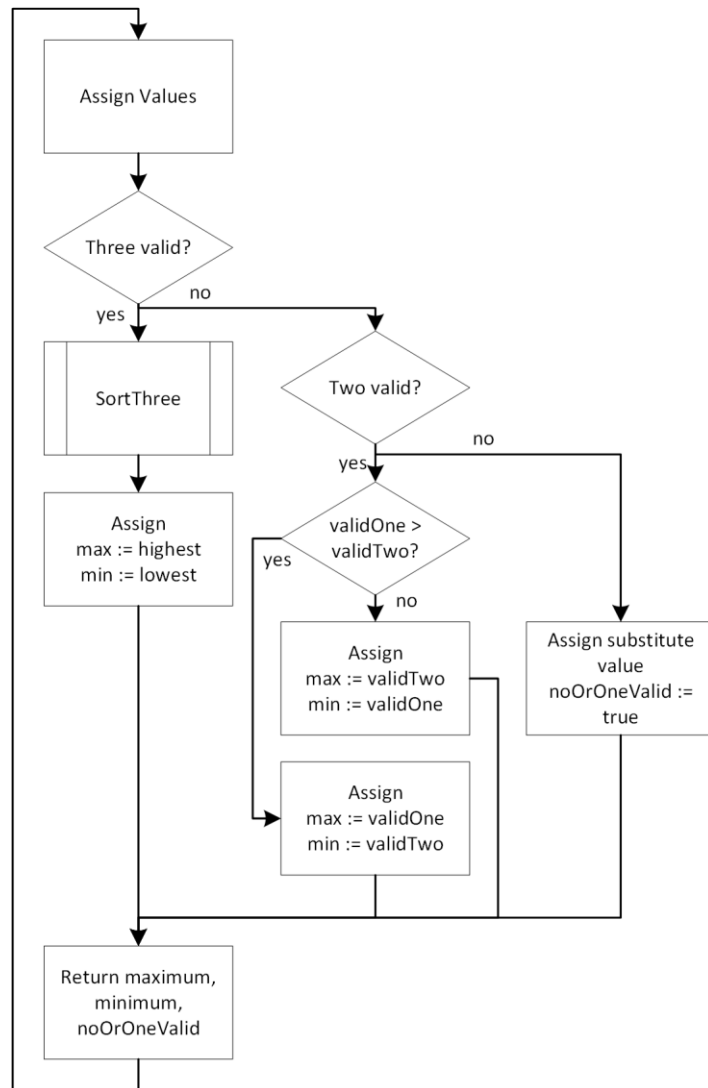
Input parameter

Identifier	Data type	Description
in1	DInt	Input 1.
in2	DInt	Input 2.
in3	DInt	Input 3.
in1Valid	Bool	True, if input 1 is valid.
in2Valid	Bool	True, if input 2 is valid.
in3Valid	Bool	True, if input 3 is valid.
substituteValue	DInt	Substitute value, if less than two inputs are valid.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
maximum	DInt	Output of maximum Median for two valid or median for three valid output of minimum Median for two valid or median for three valid inputs.
minimum	DInt	Output of minimum Median for two valid or median for three valid inputs.
noOrOneValid	Bool	Output true, if no or only one variable is valid, medianMax and medianMin will set to 0.

Functional description

**NOTICE**

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE

The enable input (EN) must not be wired for F-modules.

**WARNING**

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	3,3

4 Program blocks

CPU-Type	runtime [ms]
1510 SP F-1 PN	1,8
1511 F-1 PN / 1511 TF-1 PN	0,9
1515 F-2 PN / 1515 TF-2 PN	0,5
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

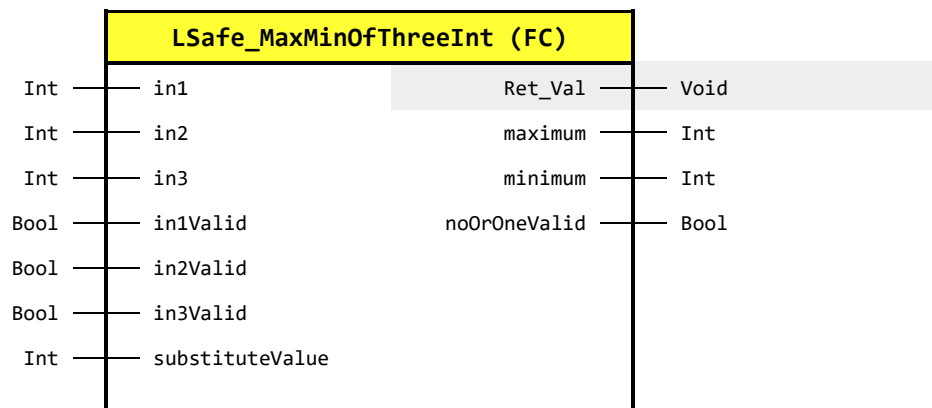
Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.2.6 LSafe_MaxMinOfThreeInt (SafetyFC / V1.0.0)

Author: Siemens

Short description

The block returns the maximum and minimum value for two or three valid integer variables. If less than two variables are valid, the substitute value is returned.

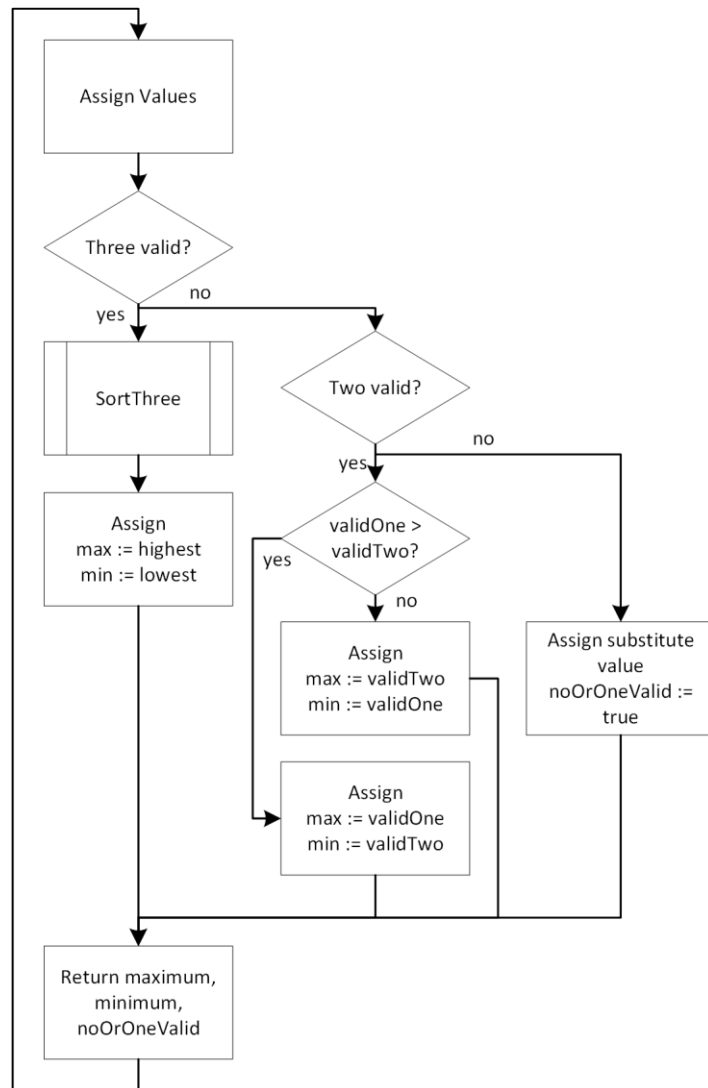
Interface description**Block interface****Input parameter**

Identifier	Data type	Description
in1	Int	Input 1.
in2	Int	Input 2.
in3	Int	Input 3.
in1Valid	Bool	True, if input 1 is valid.
in2Valid	Bool	True, if input 2 is valid.
in3Valid	Bool	True, if input 3 is valid.
substituteValue	Int	Substitute value, if less than two inputs are valid.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
maximum	Int	Output of maximum Median for two valid or median for three valid output of minimum Median for two valid or median for three valid inputs.
minimum	Int	Output of minimum Median for two valid or median for three valid inputs.
noOrOneValid	Bool	Output true, if no or only one variable is valid, medianMax and medianMin will set to 0.

Functional description



NOTICE

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE

The enable input (EN) must not be wired for F-modules.



WARNING

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	3,0
1510 SP F-1 PN	1,8
1511 F-1 PN / 1511 TF-1 PN	0,9
1515 F-2 PN / 1515 TF-2 PN	0,5
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

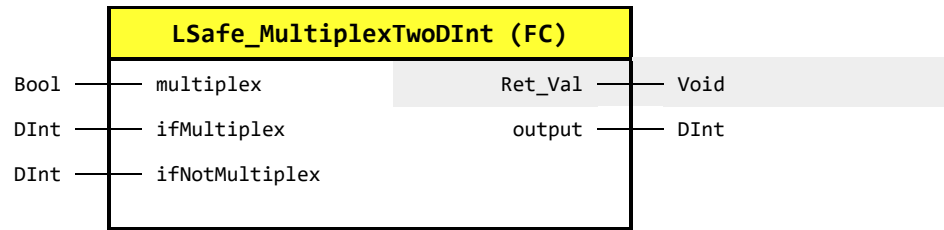
4.2.7 LSafe_MultiplexTwoDInt (SafetyFC / V1.0.0)

Short description

The block uses a boolean input to decide which of the two DoubleInteger - input values is assigned to the output value.

Interface description

Block interface



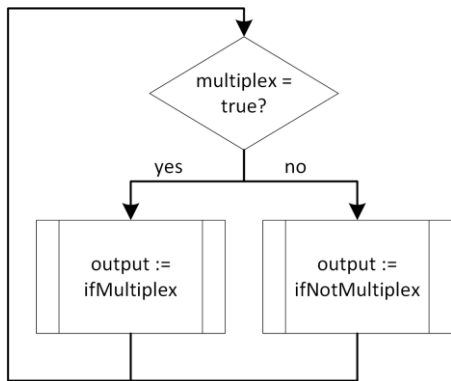
Input parameter

Identifier	Data type	Description
multiplex	Bool	Decides, what input will be assigned to the output. If true, ifMultiplex will be assigned to the output if false ifNotMultiplex.
ifMultiplex	DInt	First input, if multiplex = true, this will be assigned to the output.
ifNotMultiplex	DInt	Second input, if multiplex = false, this will be assigned to the output.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
output	DInt	Output variable.

Functional description



NOTICE	To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.
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NOTICE	The enable input (EN) must not be wired for F-modules.
--------	--

**WARNING**

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger.
Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	1,4
1510 SP F-1 PN	0,8
1511 F-1 PN / 1511 TF-1 PN	0,7
1515 F-2 PN / 1515 TF-2 PN	0,5
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First

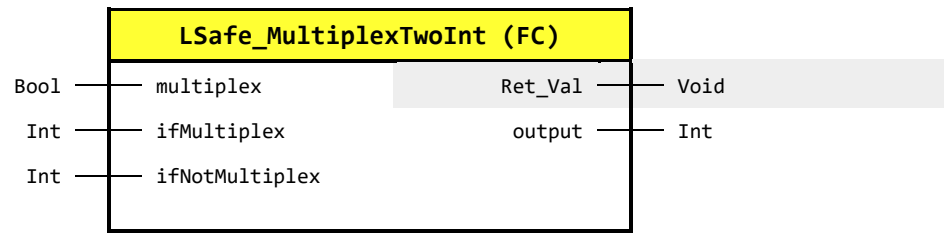
4.2.8 LSafe_MultiplexTwoInt (SafetyFC / V1.0.0)

Short description

The block uses a boolean input to decide which of the two integer input values is assigned to the output value. If Multiplex is true, the input “ifMultiplex” is assigned, otherwise “ifNotMultiplex”.

Interface description

Block interface



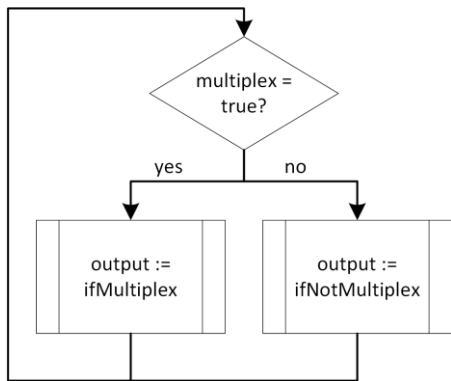
Input parameter

Identifier	Data type	Description
multiplex	Bool	Decides, what input will be assigned to the output. If true, ifMultiplex will be assigned to the output if false ifNotMultiplex.
ifMultiplex	Int	First input, if multiplex = true, this will be assigned to the output.
ifNotMultiplex	Int	Second input, if multiplex = false, this will be assigned to the output.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
output	Int	Output variable.

Functional description



NOTICE

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE

The enable input (EN) must not be wired for F-modules.

**WARNING**

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger.
Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	1,4
1510 SP F-1 PN	0,8
1511 F-1 PN / 1511 TF-1 PN	0,7
1515 F-2 PN / 1515 TF-2 PN	0,4
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.2.9 LSafe_RangeDInt (SafetyFC / V1.0.0)

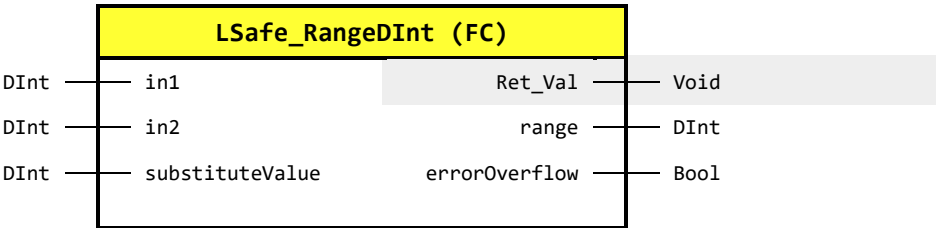
Author: Siemens

Short description

The function block calculates the distance between two double integer numbers. For this purpose, the function block evaluates whether two positive, two negative or one positive and one negative number are involved. Accordingly, it subtracts the larger number from the smaller number in the case of two positive or two negative numbers. In case of two unequal signs, both amounts are added. The output errorOverflow evaluates the ENO outputs of the SUB, ADD and ABS blocks.

Interface description

Block interface



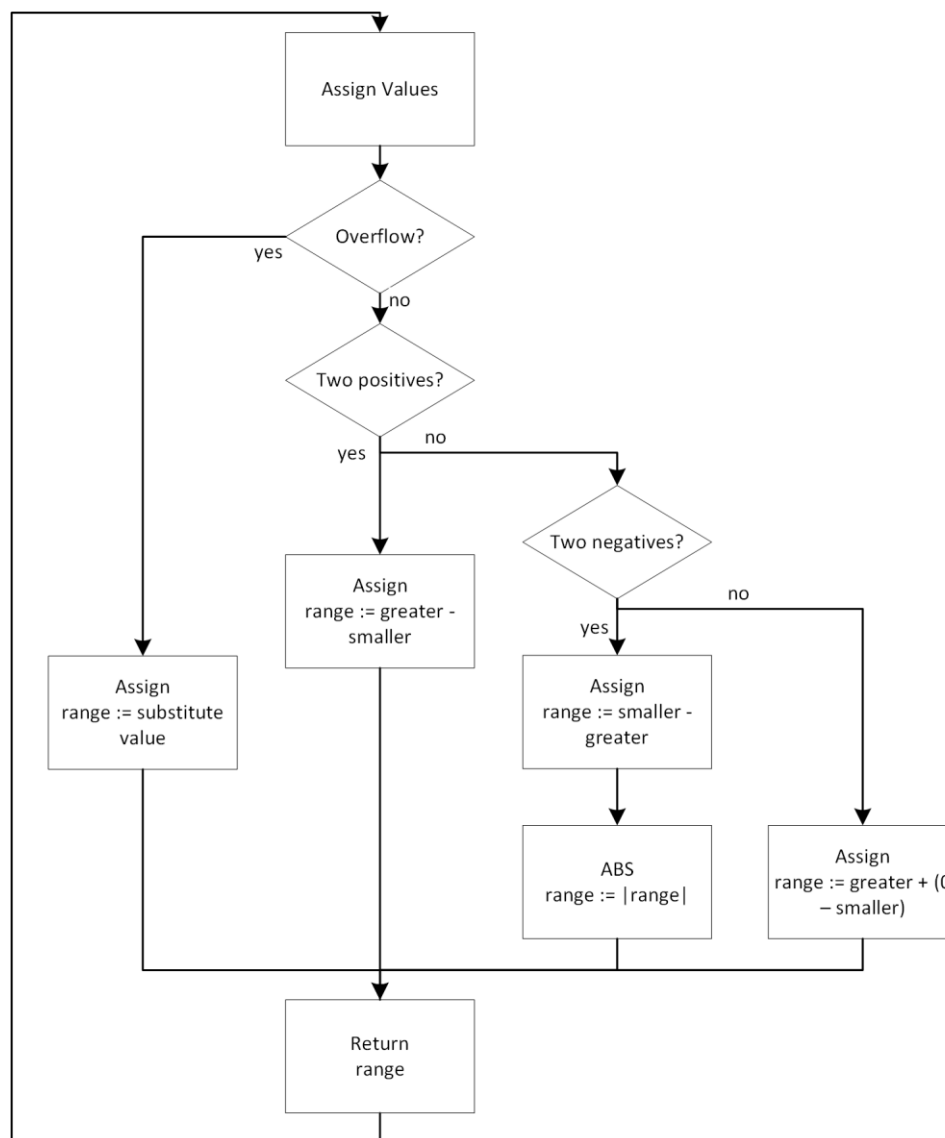
Input parameter

Identifier	Data type	Description
in1	DInt	Input 1.
in2	DInt	Input 2.
substituteValue	DInt	Substitute value, will be returned if overflow occurs.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
range	DInt	Output value, positive range between two input values, sustbitueValue if overlofError.
errorOverflow	Bool	True, when overflow error.

Functional description



NOTICE To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE The enable input (EN) must not be wired for F-modules.



WARNING

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	1,4
1510 SP F-1 PN	0,8
1511 F-1 PN / 1511 TF-1 PN	0,7
1515 F-2 PN / 1515 TF-2 PN	0,5
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.2.10 LSafe_RangeInt (SafetyFC / V1.0.0)

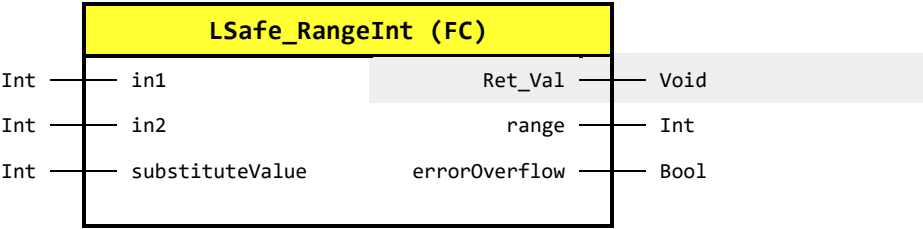
Author: Siemens

Short description

The function block calculates the distance between two integer numbers. For this purpose, the function block evaluates whether two positive, two negative or one positive and one negative number are involved. Accordingly, it subtracts the larger number from the smaller number in the case of two positive or two negative numbers. In case of two unequal signs, both amounts are added. The output errorOverflow evaluates the ENO outputs of the SUB, ADD and ABS blocks.

Interface description

Block interface



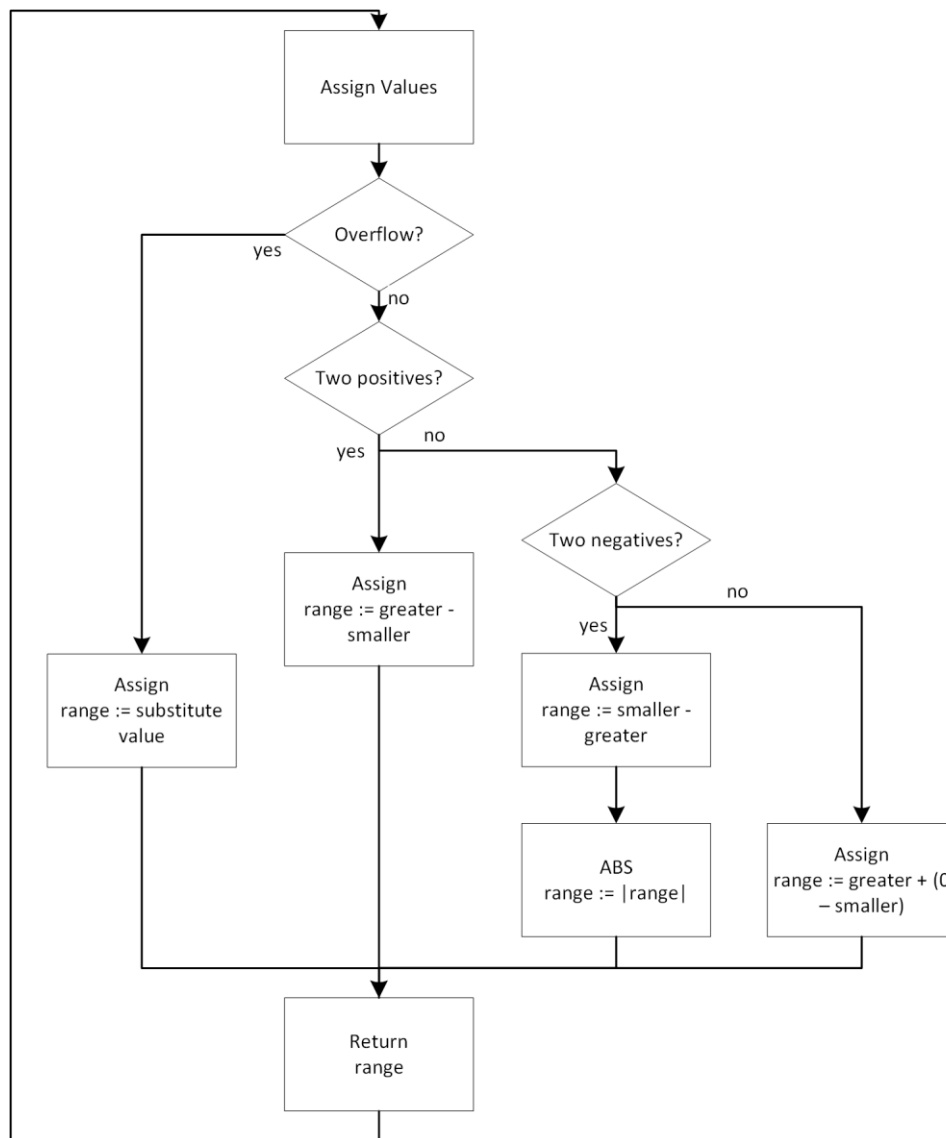
Input parameter

Identifier	Data type	Description
in1	Int	Input 1.
in2	Int	Input 2.
substituteValue	Int	Substitute value, will be returned if overflow occurs.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
range	Int	Output value, positive range between two input values, sustbitueValue if overlofError.
errorOverflow	Bool	True, when overflow error.

Functional description



NOTICE

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE

The enable input (EN) must not be wired for F-modules.



WARNING

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	1,4
1510 SP F-1 PN	0,9
1511 F-1 PN / 1511 TF-1 PN	0,8
1515 F-2 PN / 1515 TF-2 PN	0,4
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.2.11 LSafe_SortThreeDInt (SafetyFC / V1.0.0)

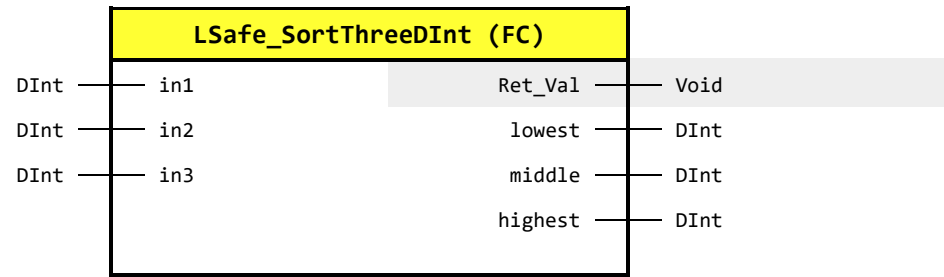
Author: Siemens

Short description

The function sorts the three DoubleInteger input variables and returns them sorted by size. A bubblesort algorithm is used for this.

Interface description

Block interface



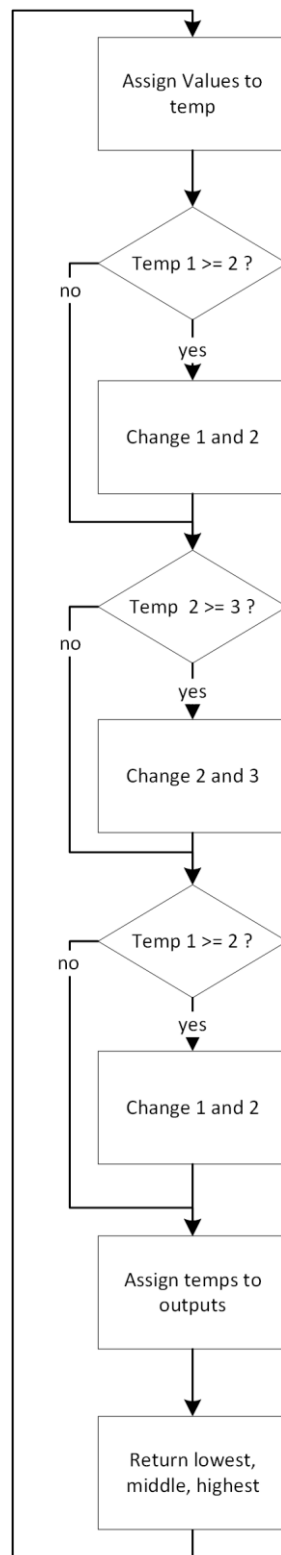
Input parameter

Identifier	Data type	Description
in1	DInt	Input 1.
in2	DInt	Input 2.
in3	DInt	Input 3.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
lowest	DInt	Output of the lowest value.
middle	DInt	Output of the middle value.
highest	DInt	Output of the highest value.

Functional description



NOTICE

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE The enable input (EN) must not be wired for F-modules.



WARNING

Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger. Please note this if you want to change the settings shown here due to individual specifications.

Typical block runtime

CPU-Type	runtime [ms]
121x FC	1,4
1510 SP F-1 PN	0,9
1511 F-1 PN / 1511 TF-1 PN	0,7
1515 F-2 PN / 1515 TF-2 PN	0,4
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

4.2.12 LSafe_SortThreeInt (SafetyFC / V1.0.0)

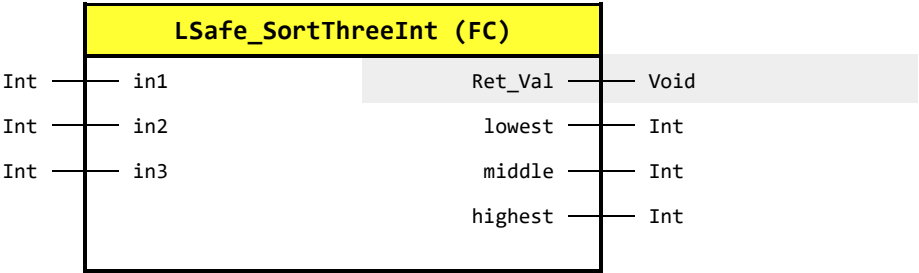
Author: Siemens

Short description

The function sorts the three integer input variables and returns them sorted by size. A bubblesort algorithm is used for this.

Interface description

Block interface



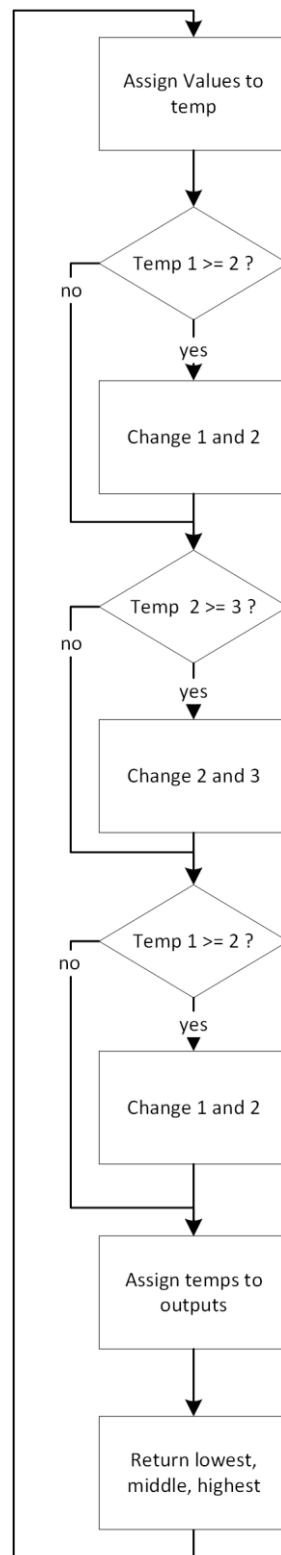
Input parameter

Identifier	Data type	Description
in1	Int	Input 1.
in2	Int	Input 2.
in3	Int	Input 3.

Output parameter

Identifier	Data type	Description
Ret_Val	Void	Void - Function has no return value
lowest	Int	Output of the lowest value.
middle	Int	Output of the middle value.
highest	Int	Output of the highest value.

Functional description



NOTICE

To achieve the desired SIL/PL level, appropriately qualified encoders and actuators must be used.

NOTICE The enable input (EN) must not be wired for F-modules.



WARNING

**Unsuitable values for safety-relevant settings can lead to the safety function not being triggered in the event of danger.
Please note this if you want to change the settings shown here due to individual specifications.**

Typical block runtime

CPU-Type	runtime [ms]
121x FC	1,4
1510 SP F-1 PN	0,9
1511 F-1 PN / 1511 TF-1 PN	0,8
1515 F-2 PN / 1515 TF-2 PN	0,4
1516 F-3 PN/DP / 1516 TF-3 PN/DP	0,3
1517 F-3 PN/DP / 1517 TF-3 PN/DP	0,1
1518 F-4 PN/DP (T, ODK bzw. MEP)	0,1

Change log

Version & Date	Change description
01.00.00 31.02.2024	Siemens Industry Online Support First released

5 Appendix

5.1 Services and support

SiePortal

The integrated platform for product selection, purchasing and support - and connection of Industry Mall and Online support. The SiePortal home page replaces the previous home pages of the Industry Mall and the Online Support Portal (SIOS) and combines them.

- **Products & Services**
In Products & Services, you can find all our offerings as previously available in Mall Catalog.
- **Support**
In Support, you can find all information helpful for resolving technical issues with our products.
- **mySiePortal**
mySiePortal collects all your personal data and processes, from your account to current orders, service requests and more. You can only see the full range of functions here after you have logged in.

You can access SiePortal via this address: sieportal.siemens.com

Technical Support

The Technical Support of Siemens Industry provides you fast and competent support regarding all technical queries with numerous tailor-made offers – ranging from basic support to individual support contracts.

Please send queries to Technical Support via Web form:
support.industry.siemens.com/cs/my/src

SITRAIN – Digital Industry Academy

We support you with our globally available training courses for industry with practical experience, innovative learning methods and a concept that's tailored to the customer's specific needs.

For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page:

siemens.com/sitrain

Industry Online Support app

You will receive optimum support wherever you are with the "Industry Online Support" app. The app is available for iOS and Android:



5.2 Links und Literatur

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to this application example https://support.industry.siemens.com/cs/ww/en/view/10000
\3\	Programming Guidelines and Programming Styleguide for SIMATIC S7-1200 and S7-1500 and WinCC (TIA Portal) https://support.industry.siemens.com/cs/ww/en/view/81318674
\4\	Libraries of PLC data types (LPD) for STEP 7 (TIA Portal) and SIMATIC S7-1200 / S7-1500 https://support.industry.siemens.com/cs/ww/en/view/109482396
\5\	Guideline on Library Handling in TIA Portal https://support.industry.siemens.com/cs/ww/en/view/109747503
\6\	Libraries in the TIA Portal https://support.industry.siemens.com/cs/ww/en/view/109738702
\7\	LaTeX project https://www.latex-project.org/
\8\	LaTeX @ Wikipedia https://de.wikipedia.org/wiki/LaTeX
\9\	LaTeX & LaTeX Mathematik @ Wikipedia https://en.wikibooks.org/wiki/LaTeX https://en.wikibooks.org/wiki/LaTeX/Mathematics
\10\	Pandoc - document render (render of mathematical formulas) https://pandoc.org/

5.3 Change log

Version & Date	Change description
V1.0.1 (10/2024)	New - Typical block runtime for CPU1516F/ 1517F added (all blocks) - Supplementary note on the validity of the TÜV report (page 4) Modification - Adaptation of short description LSafe_1oo2Allnt - Adaptation of short description LSafe_2003DI - Adaptation Short description LSafe_2oo3AIDInt - Adaptation Short description LSafe_2oo3Allnt - Adaptation Functional description LSafe_2003DI - Adaptation Short description LSafe_MonitorLimitDInt - Adaptation Short description LSafe_MonitorLimitInt - Adaptation Short description LSafe_AverageOfThreeInt - Adaptation Functional description LSafe_AverageOfThreeInt
V1.0.0 02/2024	NEW: - LSafe_1oo2Allnt / V01.00.00 • First - LSafe_2oo3AIDInt / V01.00.00 • First released - LSafe_2oo3Allnt / V01.00.00 • First - LSafe_2oo3DI / V01.00.00 • First released - LSafe_MonitorLimitDInt / V01.00.00 • First released - LSafe_MonitorLimitInt / V01.00.00 • First released - LSafe_AverageOfThreeDInt / V01.00.00 • First released - LSafe_AverageOfThreeInt / V01.00.00 • First released - LSafe_InRangeDInt / V01.00.00 • First released - LSafe_InRangeInt / V01.00.00 • First released - LSafe_MaxMinOfThreeDInt / V01.00.00 • First released - LSafe_MaxMinOfThreeInt / V01.00.00 • First released - LSafe_MultiplexTwoDInt / V01.00.00 • First - LSafe_MultiplexTwoInt / V01.00.00 • First released - LSafe_RangeDInt / V01.00.00 • First released - LSafe_RangeInt / V01.00.00

Version & Date	Change description
	- First released LSafe_SortThreeDInt / V01.00.00 - First released LSafe_SortThreeInt / V01.00.00 - First released